Name

Mathematics Extension 1 Year 11

Permutations & Combinations

Book 1

Version: 261028

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Syllabus Content

ME1-11-04 uses permutations and combinations to solve problems involving counting, ordering and probability

- Use the notation $n!$ (read as n factorial), where $n! = n(n-1)(n-2) \times ... \times 3 \times 2 \times 1$ for positive integers n
- Use $n! = n \times (n-1)!$ and the convention $0! = 1$ in calculations and to simplify algebraic expressions involving factorials
- Establish and use the multiplication principle: if a selection can be made in two stages, where there are m choices for the first stage and n choices for the second stage then there are $m \times n$ choices for the selection
- Apply the multiplication principle to explain why the number of ways of ordering n distinct objects in a straight line is $n!$
- Define a permutation as an ordered selection of some or all objects from a set of distinct objects
- Use the notation ${}^nP_r$ to represent an ordered selection of r objects from n distinct objects and observe that ${}^nP_n = n!$ and ${}^nP_0 = 1$
- Use the multiplication principle to establish that the number of ordered selections of r objects from n distinct objects is $n(n-1)(n-2) \times ... \times (n-r+1)$ and show that ${}^nP_r = n(n-1)(n-2) \times ... \times (n-r+1) = \frac{n!}{(n-r)!}$
- Solve problems involving permutations, including situations where the objects are not all distinct
- Solve problems involving permutations with restrictions on the placement of one or more objects
- Explain why the number of ways to arrange n distinct objects in a circle is $(n-1)!$
- Solve problems involving circular arrangements of distinct objects with or without restrictions on the placement of one or more objects
- Define a combination and use the notation ${}^nC_r$ or $\binom{n}{r}$ to represent the number of ways of selecting a subset of r objects from n distinct objects, where order is not important
- Establish and use the formula ${}^nC_r = \frac{n!}{r!(n-r)!}$
- Show that ${}^nC_n = {}^nC_0 = 1$ and ${}^nC_1 = {}^nC_{n-1} = n$
- Show that ${}^nC_r = {}^nC_{n-r}$, for $0 \leq r \leq n$ by selecting r objects from n distinct objects for inclusions and $n-r$ objects from n distinct objects for exclusion
- Prove ${}^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$ for $1 \leq r \leq n-1$ algebraically and using combinatorial arguments
- Solve problems involving combinations with or without restrictions on the selection of one or more objects
- Solve problems involving both permutations and combinations, including problems which require consideration of cases
- Solve probability problems involving permutations and combinations

Factorial Notation

ⓘ Factorials

Factorials let you quickly write products of descending whole numbers.

$$n! = n \times (n-1) \times (n-2) \ldots \times 2 \times 1 \quad \text{For } n \in \{0, 1, 2, 3 \ldots\}$$

⚠ Mathematicians define $0! = 1$.
⚠ You can only have factorials of whole numbers; 1.5! is undefined.

Interpret Factorial notation

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$11! = 11 \times 10 \times 9 \times 8 \ldots \times 2 \times 1$$

$$(n+2)! = (n+2) \times (n+1) \times n \times (n-1) \ldots \times 2 \times 1$$

Write out at least the first four factors of these factorials.

a 7!

b 29!

c 932!

d $(n+1)!$

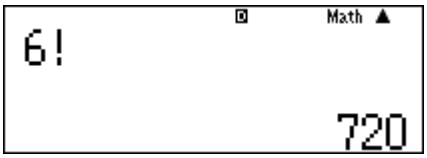
e $(n-2)!$

f $(n+r)!$

ⓘ Factorials on a Calculator

FX-82 AU

Use the $x!$ button.



FX-8200 AU

⊛ – [Probability] > [Factorial(!)]

ⓘ Recursive Definition of Factorials

$0! = 1$

$n! = n \times (n-1)!$ For $n \in \{0, 1, 2, 3 \ldots\}$

For example, $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$

$= 6 \times 5!$

ⓘ Unrolling Factorials

Simplifying factorial expressions usually involves "unrolling" them using the recursive definition.

$8! = 8 \times 7!$ (unrolling once)

$8! = 8 \times 7 \times 6!$ (unrolling twice)

$8! = 8 \times 7 \times 6 \times 5!$ (unrolling three times)

Unroll factorials a given number of times.

Unroll $5!$ once. $5! = 5 \times 4!$

Unroll $11!$ three times. $11! = 11 \times 10 \times 9 \times 8!$

Unroll $n!$ r times. $n! = n \times (n-1) \times (n-2) \ldots \times (n-r+1) \times (n-r)!$

Unroll these factorials the given number of times.

a Unroll 7! two times

b Unroll 14! three times

c Unroll 14! four times

d Unroll 800! four times

e Unroll 800! r times

f Unroll $(n+2)!$ two times.

Example Simplify factorial expressions.

$\frac{10!}{7!}$

$\frac{10!}{7!} = \frac{10 \times 9 \times 8 \times 7!}{7!}$

$= 10 \times 9 \times 8$

$= 720$

$\frac{(n+2)!}{(n-1)!}$

$\frac{(n+2)!}{(n-1)!} = \frac{(n+2)(n+1)(n)(n-1)!}{(n-1)!}$

$= (n+2)(n+1)n$

Practice

a $\frac{13!}{10!}$

1716

b $\frac{26!}{27!}$

$\frac{1}{27}$

c $\frac{(n-2)!}{(n+1)!}$

$\frac{1}{(n+1)n(n-1)}$

d $\frac{n!}{(n-r)!}$

$n(n-1)(n-2)\ldots(n-r+1)$

Example Simplify factorial expressions.

$\frac{1}{8!} - \frac{1}{10!}$

$\frac{1}{8!} - \frac{1}{10!} = \frac{10 \times 9}{10!} - \frac{1}{10!}$

$= \frac{89}{10!}$

$\frac{1}{n!} + \frac{1}{(n+1)!}$

$\frac{1}{n!} + \frac{1}{(n+1)!} = \frac{n+1}{(n+1)!} + \frac{1}{(n+1)!}$

$= \frac{n+2}{(n+1)!}$

Practice

a $\frac{1}{10!} - \frac{1}{9!}$

$-\frac{9}{10!}$

b $\frac{1}{(n-1)!} - \frac{1}{n!}$

$\frac{n-1}{n!}$

1 Use the definition of n! and methods of unrolling factorials to evaluate each expression. Do not use a calculator.

a $3!$

$3! = 3 \times 2 \times 1 = 6$

b $5!$

$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

c $1!$

$1! = 1$

d $\dfrac{15!}{14!}$

$\dfrac{15!}{14!} = \dfrac{15 \times 14!}{14!} = 15$

e $\dfrac{10!}{8! \times 2!}$

$\dfrac{10!}{8! \times 2!} = \dfrac{10 \times 9 \times 8!}{8! \times 2} = \dfrac{10 \times 9}{2} = 45$

f $\dfrac{15!}{13! \times 3!}$

$\dfrac{15!}{13! \times 3!} = \dfrac{15 \times 14 \times 13!}{13! \times 6} = \dfrac{15 \times 14}{6} = 35$

g $\dfrac{12!}{3! \times 9!}$

$\dfrac{12!}{3! \times 9!} = \dfrac{12 \times 11 \times 10 \times 9!}{6 \times 9!} = \dfrac{12 \times 11 \times 10}{6} = 220$

h $\dfrac{8!}{4! \times 4!}$

$\dfrac{8!}{4! \times 4!} = \dfrac{8 \times 7 \times 6 \times 5 \times 4!}{24 \times 4!} = \dfrac{8 \times 7 \times 6 \times 5}{24} = 70$

2 Use the factorial button n! on your calculator to evaluate each expression.

a $7!$

5040

b $0!$

1

c $10!$

3 628 800

d $\dfrac{8!}{3!}$

6720

e $\dfrac{10!}{5! \times 3! \times 2!}$

2520

f $\dfrac{12!}{2! \times 3! \times 4! \times 5!}$

13 860

3 If $f(x) = x^6$, find expressions for:

a $f'(x)$

$6x^5$

b $f''(x)$

$30x^4$

c $f'''(x)$

$120x^3$

d $f^{(4)}(x)$

$360x^2$

e $f^{(5)}(x)$

$720x$

f $f^{(6)}(x)$

720

4 Simplify by unrolling factorials appropriately:

a $\dfrac{n!}{(n-1)!}$

$\dfrac{n(n-1)!}{(n-1)!} = n$

b $n \times (n-1)!$

$n!$

c $\dfrac{n(n-1)!}{n!}$

$\dfrac{n(n-1)!}{n(n-1)!} = 1$

d $\dfrac{(n+1)!}{(n-1)!}$

$\dfrac{(n+1)n(n-1)!}{(n-1)!} = n(n+1)$

e $\dfrac{(n+2)!}{n!}$

$\dfrac{(n+2)(n+1)n!}{n!} = (n+1)(n+2)$

f $\dfrac{(n-2)!}{n!}$

$\dfrac{(n-2)!}{n(n-1)(n-2)!} = \dfrac{1}{n(n-1)}$

5 Simplify:

a $\dfrac{(n-2)!(n-1)!}{n!(n-3)!}$

$\dfrac{(n-2)!\,(n-1)!}{n(n-1)!\,(n-3)!} = \dfrac{(n-2)!}{n(n-3)!} = \dfrac{(n-2)(n-3)!}{n(n-3)!} = \dfrac{n-2}{n}$

b $\dfrac{n!(n-1)!}{(n+1)!}$

$\dfrac{n!\,(n-1)!}{(n+1)n!} = \dfrac{(n-1)!}{n+1}$

6 Simplify by factorising:

a 8! - 7!

$8(7!) - 7! = 7!\,(8-1) = 7 \times 7!$

b (n+1)! - n!

$(n+1)n! - n! = n!\,\big((n+1)-1\big) = n \times n!$

c 8! + 6!

$8 \times 7 \times 6! + 6! = 6!\,(56+1) = 57 \times 6!$

d (n+1)! + (n-1)!

$(n+1)n(n-1)! + (n-1)!$
$= (n-1)!\,\big((n+1)n+1\big)$
$= (n^2+n+1) \times (n-1)!$

e 9! + 8! + 7!

$9 \times 8 \times 7! + 8 \times 7! + 7!$
$= 7!\,(72+8+1)$
$= 9^2 \times 7!$

f (n+1)! + n! + (n-1)!

$(n+1)n(n-1)! + n(n-1)! + (n-1)!$
$= (n-1)!\,\big((n+1)n+n+1\big)$
$= (n+1)^2 \times (n-1)!$

7 Write each expression as a single fraction.

a $\dfrac{1}{n!} + \dfrac{1}{(n-1)!}$

$\dfrac{1}{n!} + \dfrac{n}{n!} = \dfrac{1+n}{n!}$

b $\dfrac{1}{n!} - \dfrac{1}{(n+1)!}$

$\dfrac{n+1}{(n+1)!} - \dfrac{1}{(n+1)!} = \dfrac{n}{(n+1)!}$

c $\dfrac{1}{(n+1)!} - \dfrac{1}{(n-1)!}$

$\dfrac{1}{(n+1)!} - \dfrac{n(n+1)}{(n+1)!}$
$= \dfrac{1-n(n+1)}{(n+1)!}$
$= \dfrac{1-n-n^2}{(n+1)!}$

8 If $f(x) = x^n$, find:

a $f'(x)$

nx^{n-1}

b $f''(x)$

$n(n-1)x^{n-2}$

c $f^{(n)}(x)$

$n!$

d $f^{(k)}(x),\ k \le n$

$n(n-1)(n-2)\ldots(n-k+1)x^{n-k}$
$= \left(\dfrac{n!}{(n-k)!}\right)x^{n-k}$

9 If $f(x) = \frac{1}{x}$, find:

a $f'(x)$

$-1 \times x^{-2}$

b $f''(x)$

$(-1)^2 \times 2! \times x^{-3}$

c $f^{(5)}(x)$

$(-1)^5 \times 5! \times x^{-6}$

d $f^{(n)}(x)$

$(-1)^n \times n! \times x^{-(n+1)}$

10

a Show that $k \times k! = (k+1)! - k!$

$RHS = (k+1)! - k!$

$= (k+1)k! - k!$

$= k!\big((k+1) - 1\big)$

$= k \times k!$

b Hence by considering each individual term as a difference of two terms, sum the series

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \ldots + n \times n!$$

$1 \times 1! = 2! - 1!$

$2 \times 2! = 3! - 2!$

$3 \times 3! = 4! - 3!$

$n \times n! = (n+1)! - n!$

so, $(2! - 1!) + (3! - 2!) + (4! - 3!) \ldots (n+1)! - n!$

Every term cancels except $(n+1)! - 1!$

11 Find the largest power of 2 that is a factor of 10!

Consider prime factorisation of each factor in 10!

$10! = 1 \times 2 \times 3 \times 2^2 \times 5 \times 2(3) \times 7 \times 2^3 \times 3^2 \times 2(5) = 2^8 \times 3^4 \times 5^2 \times 7$

$\therefore 2^8$

12 Find the largest power of 10 that is a factor of 10!

$10 = 2(5)$

$10! = 2^8 \times 3^4 \times 5^2 \times 7$

You can make 2 lots of (2×5), so 10^2 is the highest factor

13 Find the largest power of 5 that is a factor of 100!

Consider multiples of 5 in 100!:

There are 20 multiples of 5: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100

Four of these are multiples of 25: 25, 50, 75, 100

20 multiples of 5 contribute 20 factors of 5.

4 multiples of 25 contribute 4 additional factors of 5.

Total = 20 + 4 = 24 factors of 5.

$\therefore 5^{24}$ is the highest factor.

Ordered Selections with Repetition

ⓘ Fundamental Counting Principle

If one event can occur in m ways and another event can occur in n ways,
then both events can occur in $m \times n$ ways.

This is also called the **multiplication principle**. You have used this in probability calculations.

Calculate total number of outcomes.

a Timmy has two shirts and three trousers.

How many outfit combinations can he have?

$2 \times 3 = 6$

b Tammy has three shirts and five skirts.

How many outfit combinations can she have?

c A restaurant offers a three-course set menu.

Starter: 3 options | Main: 5 options | Dessert: 2 options

How many meal combinations are there?

d A student ID has six characters. It starts with an English letter and has five digits.

How many students can this ID system support?

e A dice is rolled and a coin is flipped.

How many possible outcomes are there?

f Two dice are rolled and a coin is flipped.

How many possible outcomes are there?

The "Box Method"

Visualise counting problems by placing objects into compartments.

This helps you organise complex counting questions. Think of each compartment as a separate choice, then work out how many ways you can fill each one.

1. Set up boxes in the order of your choices.
2. Determine how many ways you can fill each box.
3. Multiply all the numbers together.

Ordered Selections with Repetition

Counting ordered selections with repetition uses repeated multiplication, so you can calculate them quickly with powers.

Example Count total number of ordered selections with repetition.

The number plate on a car has 2 letters, followed by 4 numbers.
How many different number plates of this type are possible?

L	L	N	N	N	N
26	26	10	10	10	10

$26^2 \times 10^4 = 6\,760\,000$

The digits 1 to 9 are used, with repetition, to make a five-digit even number such that it alternates between even and odd numbers.
How many different numbers are possible?

E	O	E	O	E
4	5	4	5	4

$4^3 \times 5^2 = 1600$

Practice

How many five-letter words can be formed where the second and fourth letters are vowels, and the other three letters are consonants?

$5^2 \times 21^3$

A serial code involves 5 letters at the start and then 15 numbers. How many different serial codes are possible?

$26^5 \times 10^{15}$

Key Ideas

1. You can arrange r items from n possibilities (reusing allowed) in …… ways.

1 A password has 4 letters. How many passwords are possible?

L L L L

26 26 26 26

$26^4 = 456976$

2 A motorcycle number plate is made up of 2 letters followed by 2 numbers.
How many number plates of this type are available?

L L N N

26 26 10 10

$26^2 \times 10^2 = 67600$

3 A password has 5 letters followed by 4 numbers. If I could use any letter of the alphabet and any number, how many different passwords could be formed?
Leave your answer in index form.

L L L L L N N N N

26 26 26 26 26 10 10 10 10

$26^5 \times 10^4$

4 A witness saw most of the number plate made up of 2 letters and two numbers on a getaway car except for the first letter and the last number. How many different cars do the police need to check in order to find this car?

? L N ?

26 1 1 10

$26 \times 10 = 260$

5 A certain brand of computer has a serial number made up of 10 letters then 15 numbers. How many computers with this type of serial number can be made? Leave your answer in index form.

$26^{10} \times 10^{15}$

6 Victoria has 4-digit postcodes starting with 3.
How many different postcodes are available in Victoria?

3 N N N

1 10 10 10

$10^3 = 1000$

7 A country town has telephone numbers starting with 63 followed by any 6 other numbers from 0 to 9. How many telephone numbers are possible in this town?

6 3 N N N N N N

1 1 10 10 10 10 10 10

$10^6 = 1000000$

8 Jarred has 12 tops, 5 pairs of jeans and 5 pairs of shoes in his wardrobe.
If he chooses a top, a pair of jeans and shoes at random, how many combinations are possible?

$12 \times 5 \times 5 = 300$

9 A car manufacturer produces cars in 8 different colours, with either manual or automatic gear transmission, and 4 different types of wheels. How many different combinations can it produce?

$8 \times 2 \times 4 = 64$

10 A city has a population of 3 500 000. How many digits should its telephone numbers have so that every person can have one?

Need $10^n \geq 3500000$

$n = 7$

11 A manufacturer of computer parts puts a serial number on each part, consisting of 3 letters, 4 numbers, then 4 letters. The number of parts sold is estimated to be 5 million. Will there be enough combinations on this serial number to cope with these sales?

$26^7 \times 10^4 > 5\,000\,000$

yes

12 A family of 5 people each choose a flavour of ice cream from vanilla, strawberry and chocolate. In how many ways can this happen?

$3^5 = 243$

13 How many three-digit numbers can be written down from the digits 0 to 9?
The digits need not be distinct.

Three-digit numbers cannot start with 0

N N N

9 10 10

$9 \times 10^2 = 900$

14 One hundred people each buy one ticket in a lottery. Tickets are selected one by one at random from a barrel and then replaced between selections. How many ways can the first three places be awarded?

P P P

100 100 100

$100^3 = 1000000$

15 A computer sends a string of ten binary digits, that is, each symbol can only be 0 or 1.
How many such ten-digit strings are possible?

B B B B B B B B B B

2 2 2 2 2 2 2 2 2 2

$2^{10} = 1024$

16

a If you toss a coin and roll a die, how many outcomes are possible?

C D

2 6

$2 \times 6 = 12$

b If you toss two coins and roll three dice, how many outcomes are possible?

C C D D D

2 2 6 6 6

$2^2 \times 6^3 = 864$

17 Users of automatic teller machines are required to enter a four-digit PIN. Find how many PINs:

a are possible,

$10^4 = 10000$

b consist of four distinct digits,

$10 \times 9 \times 8 \times 7 = 5040$

c consist of odd digits only,

$5^4 = 625$

d start and end with the same digit.

$10 \times 10 \times 10 \times 1 = 1000$

18 In Sydney, landline phone numbers, ignoring the area code, consist of eight digits.
Let us consider those starting with the digit 8 or 9.

a How many such phone numbers are possible?

8 or 9 N N N N N N N

2 10 10 10 10 10 10 10

2×10^7

b How many of these end in an odd number?

5 odd digits: 1, 3, 5, 7, 9

8 or 9 N N N N N N O

2 10 10 10 10 10 10 5

$2 \times 10^6 \times 5 = 10^7$

c How many consist of odd digits only?

First digit must be 9

9 O O O O O O O

1 5 5 5 5 5 5 5

$5^7 = 78125$

d How many are there that do not contain a zero, and in which the consecutive digits alternate between odd and even? (Hint: you need to consider 2 cases)

Starting with 8

9 E O E O E O E

1 4 5 4 5 4 5 4

$4^4 \times 5^3 = 256 \times 125 = 32000$

Starting with 9

8 O E O E O E O

1 5 4 5 4 5 4 5

$5^4 \times 4^3 = 625 \times 64 = 40000$

Total: 72 000

Ordered Selections without Repetition (Permutations)

Investigation Ordered selections without repetition.

How many ways can you arrange 4 people in a line?

Let the four people be A B C D.

Method 1: List them out systematically

A … … … … B … … … … C … … … … D … … … …

Each of these rearrangements is called a p……

Method 2: Box method.

How many ways can you choose the first person in the queue?
How many ways can you choose the second person?
How many ways can you choose the third person?
How many ways can you choose the fourth person?

1st	2nd	3rd	4th

Total number of ways: ……

This is an example of making an **ordered selection without repetition.**

For ordered selections without repetition with n elements,

The answer is $n \times (n-1) \times (n-2) \times (n-3) \ldots \times 2 \times 1$.

Permutations

A **permutation** is an **ordered** arrangement of distinct elements from a set without repetition.

Use factorials permutations, since available objects decrease by 1 at each stage.

The number of ways to arrange n distinct objects in a line is $n!$

Example Count total number of ordered selections without repetition.

How many ways are there to arrange 4 different books on a shelf?

B1	B2	B3	B4
4	3	2	1

$4 \times 3 \times 2 \times 1 = 4! = 24$

How many ways can the letters in MATHS be arranged?

L1	L2	L3	L4	L5
5	4	3	2	1

$5 \times 4 \times 3 \times 2 \times 1 = 5! = 120$

Practice

a How many ways can 6 different students stand in a line?

$6! = 720$

b How many ways can the letters in GENIUS be arranged?

$6! = 720$

c How many ways can 20 people form a queue?

$2.432902008 \times 10^{18}$

d How many ways can 10 people form a queue, if Maddie must be first?

$9! = 362\,880$

Key Ideas

1. You can arrange n elements in a line without repetition in …… ways.

Investigation Ordered Selections without Repetition chosen from a Larger Set

How many ways can a club of 16 members select a committee consisting of a president, vice president, treasurer and secretary?

P	VP	T	S

Would 16! be the correct way to answer this? ……

Consider that:

$$16! = 16 \times 15 \times 14 \times 13 \times 12!$$

Therefore, by rearranging, we can express:

$16 \times 15 \times 14 \times 13 =$

How many ways can a club of 20 members select a committee consisting of a president, vice president, treasurer and secretary?

P	VP	T	S

How many ways can a club of 20 members select a committee consisting of a president, vice president, treasurer, secretary, and janitor?

P	VP	T	S	J

How many ways can a club of n members select a committee consisting of r positions?

ⓘ Permutations of Subsets

To arrange r objects from a set of n objects:

$$^nP_r = \frac{n!}{(n-r)!}$$

⚠ $^nP_n = n!$

⚠ $^nP_0 = 1$

Interpret Permutation notation.

Write an expression for:

a $^6P_2$

$= 6 \times 5 = \frac{6!}{4!}$

b $^{10}P_2$

c $^{10}P_3$

d $^{10}P_{10}$

e $^nP_3$

f $^{n+1}P_3$

Write the following using permutation notation.

a 9×8

$= {}^9P_2$

b $9 \times 8 \times 7$

c $7 \times 6 \times 5$

d $n(n-1)(n-2)$

e $(n-4)(n-5)$

f 5

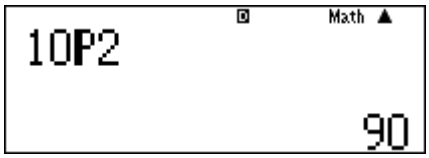
ⓘ Permutation Notation on a Calculator

FX-82 AU

SHIFT ×

FX-8200 AU

⊞ – [Probability] > [Permutation(P)]



Example Count total number of permutation arrangements.

How many two-letter codes can be formed from the letters A B C D E without repetition?

1st	2nd
5	4

$^5P_2 = 5 \times 4$
$= 20$

How many 3-digit numbers from the digits 1 to 9 are possible if you cannot repeat a digit?

1st	2nd	3rd
9	8	7

$^9P_3 = 9 \times 8 \times 7$
$= 504$

Practice

a How many four-letter words can you form from the letters Q U I C K L Y without replacing the letters?

$7P4 = 840$

b A bookshelf has space for 5 books. You have 8 different books to choose from. How many ways can you arrange 5 books on the shelf?

$8P5 = 6720$

c A committee of chairperson, secretary, and undersecretary is to be formed from a group of 15 people. How many ways can you form this committee?

$15P3 = 2730$

d A chess manager selects 3 players to play in a specific order from a team of 6. How many ways is this possible?

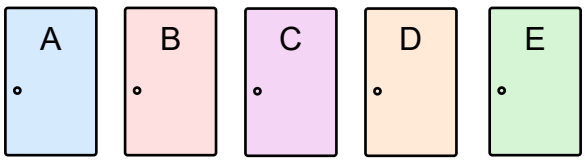
$6P3 = 120$

Key Ideas

1. A permutation refers to an o…… arrangement of d…… elements from a l…… set.
2. The notation $^{11}P_4$ means ……

Investigation

On a school trip, you have 5 single-bed hotel rooms, A B C D E, and 2 students, X Y.
How many ways can you assign the students to a room?





If position matters, consider the question as ordering in a line
(the hotel rooms themselves may not be in a straight line)

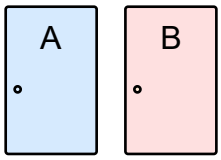
For each student, consider how many room choices they have and multiply.

You can place student X in …… possible rooms.

You can then place Student Y in …… possible rooms.

X	Y

On a school trip, you have 2 single-bed hotel rooms, A B, and 5 students, V W X Y Z.
How many ways can you assign the students to a room?





For each room, consider how many ways you can fill it.

You can fill room A with …… possible students.

You can then fill room B with …… possible students.

A	B

Key idea:

The number of ways to arrange 2 objects into 5 positions is the same as arranging 5 objects into 2 positions, both are ${}^5P_2$.

1 List all the permutations of the letters of the word DOG. How many are there?

Permutations: DOG, DGO, ODG, OGD, GOD, GDO

There are 3! = 6

2 Each of 6 people at a restaurant is given a different-coloured glass.
How many possible combinations are there?

1st 2nd 3rd 4th 5th 6th

6 5 4 3 2 1

$^6P_6 = 6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

3 A mountain trail has room for only one person at a time.
If 12 people are waiting at the bottom of the trail and are picked at random to start out,
in how many ways can this happen?

1st 2nd 3rd ... 12th

12 11 10 ... 1

$^{12}P_{12} = 12! = 479001600$

4 A dog walker has 5 dogs and 5 leashes.
In how many different ways is it possible to put a leash on each dog?

1st 2nd 3rd 4th 5th

5 4 3 2 1

$^5P_5 = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

5 A row of seats in a theatre seats 8 people.
In how many ways could a group of 8 friends be seated at random in this row?

1st 2nd 3rd 4th 5th 6th 7th 8th

8 7 6 5 4 3 2 1

$^8P_8 = 8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40320$

6 A group of 7 people line up to do karaoke.
If they are each given a song at random to sing, how many possible outcomes are there?

1st 2nd 3rd 4th 5th 6th 7th

7 6 5 4 3 2 1

$^7P_7 = 7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$

7 Find how many four-digit numbers can be formed using the digits 5, 6, 7, 8 and 9 if:

a no digit is to be repeated,

1st 2nd 3rd 4th

5 4 3 2

$^5P_4 = 5 \times 4 \times 3 \times 2 = 120$

b any of the digits can occur more than once.

1st 2nd 3rd 4th

5 5 5 5

$5^4 = 625$

8 List all the permutations of the letters ABCD, taken two at a time.

Starting with A: AB, AC, AD

Starting with B: BA, BC, BD

Starting with C: CA, CB, CD

Starting with D: DA, DB, DC

There are 4P3 = 12

9 Evaluate each permutation.

a $^{6}P_{3}$ 120	**b** $^{5}P_{2}$ 20	**c** $^{8}P_{3}$ 336
d $^{10}P_{7}$ 604 800	**e** $^{9}P_{6}$ 60 480	**f** $^{7}P_{5}$ 2520

10 Find how many arrangements of the letters of the word FRIEND are possible if the letters are taken:

a six at a time.

1st 2nd 3rd 4th 5th 6th

6 5 4 3 2 1

$^{6}P_{6} = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

b four at a time,

1st 2nd 3rd 4th

6 5 4 3

$^{6}P_{4} = 6 \times 5 \times 4 \times 3 = 360$

11 In how many ways can seven people be seated in a row of five different chairs?

1st 2nd 3rd 4th 5th

7 6 5 4 3

$^{7}P_{5} = 7 \times 6 \times 5 \times 4 \times 3 = 2520$

12 Answer the following using the notation $^{n}P_{r}$

a From a group of 10, three people line up to buy tickets. How many ways can this happen?

$^{10}P_{3}$

b Five cards are each labelled uniquely with one of the digits 1, 2, 3, 4, 5. Three of the five cards are placed down in a row. How many ways can the cards be arranged?

$^{5}P_{3}$

c One hundred people each buy one ticket in a lottery. How many ways can the first three places be awarded?

$^{100}P_{3}$

13 There are 12 swimmers in a race.

a In how many ways could they finish?

$^{12}P_{12} = 12! = 479001600$

b In how many ways could they come in first, second and third?

$^{12}P_3 = 12 \times 11 \times 10 = 1320$

14 A set of 26 cards, each with a different letter of the alphabet, is placed into a hat and cards drawn out at random without replacement. Find the number of 'words' possible if selecting

a 2 cards

$^{26}P_2 = 650$

b 3 cards

$^{26}P_3 = 15600$

c 4 cards

$^{26}P_4 = 358800$

d 5 cards

$^{26}P_5 = 7893600$

15

a How many arrangements of the letters A, B, C and D are possible if no letter can be used twice?

$^4P_4 = 24$

b How many arrangements of any 3 of these letters are possible?

$^4P_3 = 24$

c Explain why the answers to part a and b are the same.

$4P4 = 4 \times 3 \times 2 \times 1$

$4P3 = 4 \times 3 \times 2$

So, the two are equal.

Arranging 3 from 4 is the same as arranging all 4, because once you've arranged 3 letters, you have no choice for what the fourth letter is.

16 If $^8P_{\mathrm{r}} = 336$, find the value of r.

$\frac{8!}{(8-r)!} = 336$

By guess and check, $r = 3$

(there is no inverse operation for factorials)

17 Explain why $n!$ For $n \in \{2, 3, 4, 5 \ldots\}$ is even.

For numbers 2 or greater, $n!$ will always have a factor of 2, so must be even

18 Show algebraically that ${}^{n}P_{n} = {}^{n}P_{n-1} = n!$

$${}^{n}P_{n} = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$$

$${}^{n}P_{n-1} = \frac{n!}{(n-(n-1))!} = \frac{n!}{1!} = n!$$

19 Complete the sentences to give a *combinatorial argument* for why ${}^{n}P_{n-1} = n!$.

${}^{n}P_{n-1}$ means to arrange …… objects into …… positions. You can do this by choosing which one object to leave out, then arranging the remaining objects. Using the multiplication principle, this is …… × …… = $n!$

$n \mid n-1 \mid n \mid (n-1)!$

20 If $7 \times {}^{2n}P_{n} = 4 \times {}^{2n+1}P_{n}$, find n.

$$7 \times \frac{(2n)!}{n!} = 4 \times \frac{(2n+1)!}{(n+1)!}$$

$$7 \times \frac{(2n)!}{n!} = 4 \times \frac{(2n+1)(2n)!}{(n+1)\,n!}$$

Divide both sides by $\frac{(2n)!}{n!}$

$$7 = \frac{4(2n+1)}{n+1}$$

$$7(n+1) = 4(2n+1)$$

$$7n + 7 = 8n + 4$$

$$3 = n$$

21 Prove that ${}^{n+1}P_{r} = {}^{n}P_{r} + r \times {}^{n}P_{r-1}$

$$LHS = {}^{n+1}P_{r} = \frac{(n+1)!}{(n+1-r)!}$$

$$RHS = {}^{n}P_{r} + r \times {}^{n}P_{r-1}$$

$$= \frac{n!}{(n-r)!} + r \times \frac{n!}{(n-r+1)!}$$

$$= \frac{n!}{(n-r)!} + r \times \frac{n!}{(n-r+1)(n-r)!}$$

$$= \frac{n!}{(n-r)!} \times \left[1 + \frac{r}{n-r+1}\right]$$

$$= \frac{n!}{(n-r)!} \times \left[\frac{n-r+1+r}{n-r+1}\right]$$

$$= \frac{n!}{(n-r)!} \times \frac{n+1}{n+1-r}$$

$$= (n+1) \times \frac{n!}{(n+1-r)!}$$

$$= \frac{(n+1)!}{(n+1-r)!}$$

$$= LHS$$

Permutations with Restrictions

ⓘ Dealing with Restrictions

When counting ordered selections, deal with any restrictions first.

Set up boxes in the order of the selections, **not** the physical order of the objects.

Example Count total number of arrangements involving restrictions.

A five-digit number is formed from the digits 0 to 9 without repeats. How many numbers are possible?

1st	2nd	3rd	4th	5th
9	9	8	7	6

$9 \times 9 \times 8 \times 7 \times 6 = 27\,216$

Six people form two queues, each with three people. Alice will only stand in the left-hand queue, and Bob will only stand in the right-hand queue. In how many ways can the two queues be formed?

A	B	4 others
3	3	4!

$3 \times 3 \times 4! = 216$

Practice

A three-digit number is formed from the digits 5 to 9 without repetition. How many numbers greater than 700 are possible?

$3 \times 4 \times 3 = 36$

A race has 8 runners wearing numbered bibs from 1 to 8. The top 3 finishers get medals (gold, silver, bronze). How many possible medal outcomes have runner 5, 6, 7, or 8 winning gold?

$4 \times 7 \times 6 = 168$ ways

Six people are going on a hike in two teams of three. Each team needs a navigator, medic, and tent-carrier. Alison is trained as a navigator so would prefer that role. Beth and Alison are besties so want to go together. How many ways can the two teams be formed?

$2 \times 2 \times 4! = 96$ ways

Eight people form two queues, each with four people. Albert will only stand in the left-hand queue, Beth only in the right-hand queue, and Charles and Diana insist on standing in the same queue. In how many ways can the two queues be formed?

$4 \times 4 \times 6 \times 2 \times 4! = 4608$

1 A school talent quest has 11 performers and each one is randomly given the order in which to perform.

a In how many ways can the order of performances be arranged?

1st 2nd ... 11th

11 10 ... 1

$^{11}P_{11} = 11! = 39916800$

b If one performer is chosen to perform first, in how many ways can the others be arranged?

1st 2nd ... 11th

1 10 ... 1

$^{10}P_{10} = 10! = 3628800$

2 A group of 6 friends sit in the same row at a concert.

a In how many different ways can they arrange themselves?

1st 2nd ... 6th

6 5 ... 1

$^{6}P_{6} = 6! = 720$

b If one friend must sit on the centre aisle, in how many ways can they be arranged?

Centre aisle fixed, arrange remaining 5

Centre Others

1 5!

5! = 120

3 A random 3-digit number is made from cards containing the numbers 0 to 9.

a In how many ways can this be done if the cards cannot be used more than once and 0 cannot be the first number?

First digit: 1-9 (9 choices)

Second digit: 0-9 except the one used for the first digit (9 choices)

Third digit: 8 choices left

1st 2nd 3rd

9 9 8

9 × 9 × 8 = 648

b How many numbers over 400 can be made?

1st 2nd 3rd

6 9 8

First digit: 4-9 (6 choices)

Remaining digits to choose from: 9, then 8

6 × 9 × 8 = 432

c How many numbers less than 300 can be made?

1st 2nd 3rd

2 9 8

First digit: 1 or 2 (2 choices) Remaining digits to choose from: 9, then 8

2 × 9 × 8 = 144

4 A set of 5 cards, each with a number from 1 to 5 on it, is placed in a box and 2 cards are drawn out at random to make a number. Find how many possible numbers there are:

a Altogether

${}^5P_2 = 5 \times 4 = 20$

b of numbers greater than 50

first digit must be 5
$1 \times 4 = 4$

c of odd numbers

The last digit must be 1, 3, or 5.
last 1st

3 4

$3 \times 4 = 12$

d of even numbers

There last digit must be 2 or 4.
last 1st

2 4

$2 \times 4 = 8$

5 A 4-digit number is to be selected at random from the numbers 0 to 9 with a non-zero first digit and no repetition. How many arrangements are there:

a In total?

1st 2nd 3rd 4th

9 9 8 7

First digit: 1-9 Remaining choices: 9, 8, 7
$9 \times 9 \times 8 \times 7 = 4536$

b over 6000?

1st 2nd 3rd 4th

4 9 8 7

First digit: 6, 7, 8, 9 (4 choices)
$4 \times 9 \times 8 \times 7 = 2016$

c less than 8000?

1st 2nd 3rd 4th

7 9 8 7

First digit: 1-7 (7 choices)
$7 \times 9 \times 8 \times 7 = 3528$

6 The numbers 1, 2, 3, 4 and 5 are arranged in a line. How many arrangements are possible if

a there is no restriction

${}^5P_5 = 5! = 120$

b the number is less than 30000

first digit is 1 or 2
1st 2nd 3rd 4th 5th

2 4 3 2 1

$2 \times 4! = 48$

c the number is greater than 20000

Numbers > 20000: first digit is 2, 3, 4, or 5
1st 2nd 3rd 4th 5th

4 4 3 2 1

$4 \times 4! = 96$

d the number is odd

Odd numbers: last digit is 1, 3, or 5
Last 1st 2nd 3rd 4th

3 4 3 2 1

$3 \times 4 \times 3 \times 2 \times 1 = 72$

e any 3 numbers are selected at random?

${}^5P_3 = 5 \times 4 \times 3 = 60$

7 A group of 5 boys and 5 girls line up outside a cinema. In how many ways can they be arranged

a with no restriction

$^{10}P_{10} = 10! = 3628800$

b if a particular girl stands in line first

Particular girl in first position, arrange remaining 9
9! = 362880

c if boys and girls alternate (with either a girl or boy in first place)?

Case 1: Girl first
G B G B G B G B G B

5 5 4 4 3 3 2 2 1 1

5! × 5! = 120 × 120 = 14400

Case 2: Boy first
B G B G B G B G B G

5 5 4 4 3 3 2 2 1 1

5! × 5! = 14400
Total: 14400 + 14400 = 28800

8 4-digit numbers are formed from the digits 1, 2, … 8, 9, without repetition.

a How many can be formed, without restriction?

$^{9}P_{4} = 9 \times 8 \times 7 \times 6 = 3024$

b How many of these end in 1?

4th 1st 2nd 3rd

1 8 7 6

8 × 7 × 6 = 336

c How many of these are even?

Even: last digit is 2, 4, 6, or 8
4th 1st 2nd 3rd

4 8 7 6

8 × 7 × 6 × 4 = 1344

d How many are divisible by 5?

Divisible by 5: must end in 5
4th 1st 2nd 3rd

1 8 7 6

8 × 7 × 6 = 336

e How many are greater than 7000?

Greater than 7000: first digit is 7, 8, or 9
1st 2nd 3rd 4th

3 8 7 6

3 × 8 × 7 × 6 = 1008

9 Repeat the previous question if repetitions are allowed.

a) $9^4 = 6561$

b) $9^3 = 729$

c) $9^3 \times 4 = 2916$

d) $9^3 = 729$

e) $3 \times 9^3 = 2187$

10 In Tasmania, a car licence plate consists of two letters followed by four digits. Find how many of these are possible:

a if there are no restrictions,

$26^2 \times 10^4 = 676 \times 10000$
$= 6760000$

b if there is no repetition of letters or digits,

L	L	N	N	N	N
26	25	10	9	8	7

$^{26}P_2 \times {}^{10}P_4 = 3276000$

c if the second letter is X and the third digit is 3,

2nd letter	3rd digit	1st letter	1st digit	2nd digit	4th digit
(1)	(1)	26	10	10	10

$26 \times 10^3 = 26000$

d if the letters are D and Q and the digits are 3, 6, 7 and 9.

Letters (D,Q)	Digits (3,6,7,9)
2!	4!

$2! \times 4! = 2 \times 24 = 48$

11

a In how many ways can the letters of the word NUMBER be arranged?

$6! = 720$

b How many begin with N?

1st	Remaining
N (1)	5!

$5! = 120$

c How many begin with N and end with U?

1st	6th	Middle 4
N (1)	U (1)	4!

$4! = 24$

d In how many is the N somewhere to the left of the U?

Total arrangements: $6! = 720$

In exactly half, N is left of U

$\frac{720}{2} = 360$

12

a How many five-digit numbers can be formed from the digits 2, 3, 4, 5 and 6, without repetition?

$5! = 120$

b How many of these numbers are greater than 56432?

Start with 6: arrange remaining 4 digits

1st	Remaining
6 (1)	4!

$4! = 24$

c How many of these numbers are less than 56432?

Easiest method is to do total subtract the 24 greater, and subtract 1 (since the question asks for less than, not less than or equal to)

$120 - 24 - 1 = 95$

13

a How many five-digit numbers can be formed from the digits 0, 1, 2, 3 and 4 if repetitions are not allowed?

1st (not 0) Remaining 4

4 4!

$4 \times 4! = 96$

b How many of these are odd?

Case 1: Last digit is 1, first digit can't be 0

5th 1st Middle 3

1 3 3!

$1 \times 3 \times 3! = 18$

Case 2: Last digit is 3, first digit can't be 0

5th 1st Middle 3

1 3 3!

$1 \times 3 \times 3! = 18$

Total: 36

c How many are divisible by 5?

Last digit must be 0

5th First 4

1 4!

$4! = 24$

14 The letters of the word THEORY are arranged randomly. Find the number of arrangements of 4 of these letters if the first letter is a consonant and the last letter is a vowel.

1st (consonant) Last (vowel) Middle 2

4 2 ${}^4P_2$

$4 \times 2 \times {}^4P_2 = 96$

15 In how many ways can a boat crew of eight women be arranged if three of the women can only row on the bow side and two others can only row on the stroke side?

Call bowside women B_1, B_2, B_3 and stroke side women S_1, S_2. Choose spots for these women, then fill the rest.

B_1 B_2 B_3 S_1 S_2 The rest

4 3 2 4 3 3!

$4 \times 3 \times 2 \times 4 \times 3 \times 3! = 1728$

Alternate method: arrange 3 bowside women in the 4 bowside spots; arrange 2 stroke side women in 4 stroke side spots, then arrange the 3 remaining

$4P3 \times 4P2 \times 3! = 1728$

16 A motor bike can carry three people: the driver, one passenger behind the driver and one in the sidecar. If among five people, only two can drive, in how many ways can the bike be filled?

Fill driver seat first, then fill the rest.

D The rest

2 ${}^4P_2$

$2 \times {}^4P_2 = 24$

Permutations involving Grouping, Complements, and Cases

ⓘ Grouping Principle

Restrictions usually involve grouping objects as a single “unit.”

1. Identify the “units.”
2. Arrange the units.
3. Arrange the individuals within each unit.
4. Multiply the arrangements.

Example Count using the grouping principle.

Arrange the 7 letters of READING so that all vowels (E, A, I) stay together.

1. Units: (E A I), R, D, N, G
2. Arrange units: 5!
3. Arrange within units: 3!
4. $5! \times 3! = 720$

4 boys and 4 girls form a queue at a bus stop. One boy-girl couple wishes to stand together. The other three girls wish to stand together. The other three boys don’t mind.

How many ways are there to form the queue?

1. Units: (B G), (G1 G2 G3), B1, B2, B3
2. Arrange units: 5!
3. Arrange within units: 2!, 3!
4. $5! \times 2! \times 3! = 1440$

Practice

Arrange the 8 letters of COMPUTER so that all consonants (C, M, P, T, R) stay together.

$4! \times 5! = 2880$

3 boys and 4 girls line up for a photo. In how many ways can they arrange themselves if all the boys must stand together?

$5! \times 3! = 720$

A bookshelf has 6 fiction and 4 non-fiction books. In how many ways can the books be arranged if all the fiction books must be kept together?

$5! \times 6! = 86\,400$

A theatre of 3 experienced and 5 beginners actors line up for a curtain call.
Two of the experienced actors are siblings and wish to stand together.
The beginners also wish to stand together.
The other experienced actor doesn't mind.
How many ways can they line up?

$3! \times 2! \times 5! = 1440$

ⓘ Complement Principle

Words like **“at least” “at most” “not” “excluding”** indicate it is best solved by considering the complementary situation.

1. Count total orderings.
2. Count unacceptable orderings.
3. Acceptable orderings = Total – Unacceptable.

Example Count using the complementary principle.

How many 7-letter words can be formed using the letters A, B, C, D, E, F, G if the two vowels must be separated by at least one consonant?

1. Total: 7!
2. Unacceptable orderings:
   Treat AE as a single unit.
   (AE), B, C, D, F, G
   Total unacceptable: $6! \times 2!$
3. $7! - (6!\ 2!) = 3600$

9 people are to form a committee of three positions: president, treasurer, secretary. Amy and Olivia refuse to serve in the committee together. How many ways are there of forming the committee?

1. Total: 9P3
2. Unacceptable orderings:
   Choose a position for Amy: 3
   Choose a position for Olivia: 2
   There are 7 people to fill the last position.
   Total unacceptable: $3 \times 2 \times 7$
3. $9P3 - 3 \times 2 \times 7 = 462$

Practice

8 people are to stand in a line. Georgina and Harriet do not wish to stand together. How many ways are there of arranging the 8 people?

$8! - (7!\ 2!) = 30\,240$

8 students are trying out for a relay race team with four positions: first leg, second leg, third leg, anchor. Two of the students, Marcus and Sophie, cannot both be on the team. How many ways are there of selecting the team?

$8P4 - (4 \times 3 \times 6 \times 5) = 1320$

ⓘ Considering Cases

Complex questions involve considering cases, you can either:

- Consider all non-overlapping cases and add them up.
- Consider overlapping cases and subtract the overlap.

Try to set up the minimum number of cases.

Example Count by considering cases.

A number is made from the digits 5, 6, 7, 8, 9 using each digit at most once. How many of these numbers are less than 800?

Case 1: 3-digit less than 800:
3 possibilities for first digit, then arrange 2 digits from 4.
$3 \times 4P2$ = 36

Case 2: 2-digit numbers:
5P2 = 20

Case 3: 1-digit numbers:
5P1 = 5

Total less than 800 = 36 + 20 + 5 = 61

How many 4-digit numbers can be formed using the digits 1, 2, 3, 4, 5, 6 (without repetition) such that the number is odd and greater than 4000?

Case 1: First digit is 4
Last digit must be 1, 3, 5, then arrange 2 digits from 4.
$3 \times 4P2 = 36$

Case 2: First digit is 5
Last digit must be 1 or 3, then arrange 2 digits from 4.
$2 \times 4P2 = 24$

Case 3: First digit is 6
Last digit must be 1, 3, 5, then arrange 2 digits from 4.
$3 \times 4P2 = 36$

Total = 36 + 24 + 36 = 96

Practice

How many numbers greater than 6000 can be formed using the digits 3, 4, 6, 8, 9 if no digit can be used more than once per number?

4-digit arrangements: $3 \times 4P3 = 72$
5-digit arrangements: $5! = 120$
Total: 192

In how many ways can six students and two teachers be arranged in a row if there are at least 3 students separating the teachers?
Hint: subtract unacceptable arrangements

Total ways: $8! = 40320$
No students between: consider TT as a unit: $7! \times 2! = 10080$
1 student between: consider TST as a unit: $6! \times 6 \times 2! = 8640$
2 students between: consider TSST as a unit: $5! \times 6 \times 5 \times 2! = 7200$
Total acceptable ways = 14 400

1 How many ways can the following words be arranged, if the vowels must be together?

a BOARDS

5 units: (OA), B, R, D
5! × 2! = 240

b RIO

2 units: (IO), R
2! × 2! = 4

c QUIT

3 units: (UI), Q, T
3! × 2! = 12

d TROUNCE

5 units: (OUE), T, R, N, C
5! × 3! = 720

2 How many arrangements of the word MATHS are possible, if:

a the T and H must remain together?

M, A, (TH), S
4! × 2! = 48

b the TH must remain together and in this order?

M, A, (TH), S
But don't rearrange the TH
4! = 42

3 How many ways can Andrew, Becky, Courtney, Dion and Ellie sit in a row, if Andrew and Becky sit together and Dion and Ellie sit together?

(AB), C, (DE)
3! × 2! × 2! = 24

4 In how many ways can three different Maths books, six different Science books and four different English books be placed on a shelf, if the books relating to each subject are to be kept together?

Treat each subject as 1 unit, and rearrange books within each subject.
3! (3 subjects) × 3! (rearrange maths books) × 6! (rearrange science books) × 4! (rearrange English books)
= 622 080

5 A class is asked to determine how many ways the letters of the word SOLAR can be arranged, if the first two positions cannot both be vowels.

a Jack decides to use cases (the word either starts with a vowel or it does not).
Use Jack's method to answer the question.

Case 1: First position is consonant
3 × 4 × 3! = 3 × 4 × 6 = 72
Case 2: First position is vowel, second is consonant
2 × 3 × 3! = 2 × 3 × 6 = 36
Total = 72 + 36 = 108

b Jill decides to consider the complementary situation (both first positions are vowels).
Use Jill's method to answer the question.

Total arrangements: 5! = 120
Both first positions are vowels: 2 × 1 × 3! = 12
Valid arrangements: 120 - 12 = 108

6 A family of two parents and two children are going on a car trip. Only the parents can drive.
If the father drives, then the mother will sit in the back seat where she feels safer.
The father always sits in a front seat of the car.
How many arrangements are possible, if two sit in the front and two in the back?

Case 1: Father drives
Father in driver seat, mother in back (2 back seats)
2 children fill remaining spots (1 front, 1 back)
$2 \times 2 = 4$ ways

Case 2: Mother drives
Mother in driver seat, father in front passenger
2 children in back: $2! = 2$ ways

Total: 6 ways

7 How many numbers can be written down using each digit of 789 at most once, if the result must be at least 80?

Two-digit numbers:
First digit 8: second is 7 or 9 (2 numbers)
First digit 9: second is 7 or 8 (2 numbers)
Total two-digit: 4

Three-digit numbers: $3! = 6$

Total = 4 + 6 = 10

8 Find how many arrangements of the letters of the word UNIFORM are possible:

a if the vowels must occupy the first, middle and last positions,

Vowels: U, I, O (3)
Consonants: N, F, R, M (4)
Arrange the 3 vowels in the 3 positions, then arrange the consonants

$3! \times 4! = 144$

b if the word must start with U and end with M,

Fix U at start and M at end
Arrange remaining 5

$5! = 120$

c if all the consonants must be in a group at the end of the word,

Fix (N, F, R, M) at the end.
Arrange the 3 vowels then arrange the 4 consonants.
$3! \times 4! = 144$

d if the M is somewhere to the right of the U.

M is to the right in half the arrangements
$\frac{7!}{2} = 2520$

9 Find how many arrangements of the letters of the word BEHAVING:

a end in NG,

NG fixed at end
Arrange remaining 6 letters:
$6! = 720$

b begin with three vowels,

Fix (E, A, I) at start.
Arrange 5 consonants, then arrange 3 vowels
$5! \times 3! = 720$

c have three vowels occurring together.

6 units to arrange: (EAI), B, H, V, N, G
Then rearrange within unit.
$6! \times 3! = 4320$

10 A Maths test is to consist of six questions. In how many ways can it be arranged so that:

a the easiest question is first and the hardest question is last

Fix easiest at the start and hardest at the end.
Arrange remaining 4
$4! = 24$

b the easiest and hardest questions are next to one another?

Group easiest and hardest as single unit.
Arrange 5 units and then arrange within the unit
$5! \times 2! = 240$

11 Four boys and four girls are to sit in a row. How many ways can this be done if:

a The boys and girls alternate

Case 1: BGBGBGBG
Arrange boys: 4!
Arrange girls 4!
Total: 4! × 4!

Case 2: GBGBGBGB
Arrange boys: 4!
Arrange girls 4!
Total: 4! × 4!

2 × 4! × 4! = 1152

b The boys and girls sit in distinct groups

Case 1: BBBBGGGG
Arrange boys: 4!
Arrange girls 4!
Total: 4! × 4!

Case 2: BBBBGGGG
Arrange boys: 4!
Arrange girls 4!
Total: 4! × 4!

$2 \times 4! \times 4! = 1152$

Alternate method: Consider 2 units (BBBB)(GGGG)
Arrange 2 units and arrange within unit
$2 \times 4! \times 4! = 1152$

12 Five-letter words are formed without repetition from the 8 letters of PHYSICAL.

a How many consist only of consonants?

$^6P_5 = 720$

b How many begin with P and end with S?

1st 5th Middle 3

1 1 $^6P_3$

$^6P_3 = 120$

c How many begin with a vowel?

1st Remaining 4

2 $^7P_4$

$2 \times {}^7P_4 = 1680$

d How many contain the letter Y?

Total - (words without Y)

$^8P_5 - {}^7P_5 = 4200$

e How many have the two vowels occurring next to one another?

Treat (I, A) as single unit
Fix (I, A) in the word, need 3 more consonants from 6, therefore, treat the 5-letter word as 4 units.

Choose 1 of 4 spots for (I, A),
ordered arrangement of 3 consonants from 6, then arrange (I, A)
$4 \times 6P3 \times 2 = 960$

f How many have the letter A immediately following the letter L?

Treat LA as single unit
Again treat 5-letter word as 4 units.

Choose 1 of 4 spots for LA, then order 3 more letters from 6
Do not rearrange LA

$4 \times 6P3 = 480$

13 Consider the 7-letter word INCLUDE.

a How many seven-letter words can be formed without repetition?

$7! = 5040$

b How many of these do not begin with I?

First Remaining

6 6!

$6 \times 6! = 4320$

Alternatively subtract those that begin with I from total:
$7! - 6! = 4320$

c How many end in L?

First 6 L

6! 1

$6! = 720$

d How many have the vowels and consonants alternating?

There are only 3 vowels, so the only pattern is CVCVCVC
$4! \times 3! = 144$

e How many have the C immediately following the D?

Treat DC as single unit
6 units to arrange: 6! = 720

f How many have the letters N and D separated by exactly 1 letter?

Treat (N _ D) as a single unit
Therefore the 7-letter word reduces to 5 units.

Rearrange N and D within (N _ D),
pick 1 letter to go in the middle from 5,
then arrange all units

$2 \times 5 \times 5! = 1200$

g How many have the letters N and D separated by exactly two letters?

Treat (N _ _ D) as a single unit
Therefore the 7-letter word reduces to 4 units.

Rearrange N and D within (N _ _ D)
ordered selection of middle 2 letters, arrange all units
$2 \times 5P2 \times 4! = 960$

h How many have the letters N and D separated by more than two letters?

Total – Adjacent – Separated by 1 letter – Separated by 2 letters

Number adjacent:
Treat ND as single unit.
6 units to arrange and then 2 ways to arrange ND
$6! \times 2 = 1440$

5040 – 1440 – 1200 – 960 = 1440

14 In how many ways can ten people be arranged in a line:

a without restriction,

10! = 3,628,800

b if one particular person must sit at either end,

Sit 1 person at either end, then arrange 9 others

$2 \times 9! = 725\,760$

c if two particular people must sit next to one another,

Treat the two people as one unit. Arrange 9 units and arrange the 2 within the unit.

$9! \times 2! = 725\,760$

d if neither of two particular people, A and B, can sit on either end of the row?

We will subtract the unacceptable cases:

Unacceptable cases:

Case 1: Person A on either end: $2 \times 9!$

Case 2: Person B on either end: $2 \times 9!$

We then need to subtract the overlapping cases (A and B both on the ends): $2 \times 8!$

$10! - (2 \times 9! + 2 \times 9! - 2 \times 8!) = 2\,257\,920$

15 Five boys and four girls form a queue at the cinema. There are two brothers who want to stand together, the remaining three boys wish to stand together, and the four girls don't mind where they stand. In how many ways can the queue be formed?

Units: {2 brothers}, {3 boys}, {4 girls separately} = 6 units total

Units Brothers 3 boys

6! 2! 3!

$6! \times 2! \times 3! = 720 \times 2 \times 6 = 8640$

16 Eight people are to form two queues of four. In how many ways can this be done if:

a there are no restrictions,

$8! = 40\,320$

b Jim will only stand in the left-hand queue,

Jim in left Arrange 7 others

4 7!

$4 \times 7! = 20\,160$

c Sean and Liam must stand in the same queue?

Two cases:

Sean in left Liam 6 others

4 3 6!

Sean in right Liam 6 others

4 3 6!

$2 \times 4 \times 3 \times 6! = 17\,280$

17 Five backpackers arrive in a city where there are five youth hostels.

a How many different accommodation arrangements are there if there are no restrictions on where the backpackers stay?

Each backpacker has 5 choices.

Backpacker1 Backpacker2 Backpacker3 Backpacker4 Backpacker5

5 5 5 5 5

$5^5 = 3125$

b How many different accommodation arrangements are there if each backpacker stays at a different youth hostel?

Backpacker1 Backpacker2 Backpacker3 Backpacker4 Backpacker5

5 4 3 2 1

5! = 120

c Suppose that two of the backpackers are brother and sister and wish to stay in the same youth hostel. How many different accommodation arrangements are there if the other three can go to any of the other youth hostels?

Siblings' hostel Other 3 backpackers

5 choices $4^3 = 64$

$5 \times 4^3 = 320$

18 Numbers less than 4000 are formed from the digits 1, 3, 5, 8 and 9, without repetition.

a How many such numbers are there?

1-digit: 5

2-digit: $5 \times 4 = 20$

3-digit: $5 \times 4 \times 3 = 60$

4-digit (start 1 or 3): $2 \times 4 \times 3 \times 2 = 48$

Total: $5 + 20 + 60 + 48 = 133$

b How many of them are odd?

1-digit: 4 (1, 3, 5, 9)

2-digit: $4 \times 4 = 16$

3-digit: $4 \times 4 \times 3 = 48$

4-digit starting 1 or 3:

Start 1, end odd (not 1): $3 \times 3 \times 2 = 18$

Start 3, end odd (not 3): $3 \times 3 \times 2 = 18$

Total: $4 + 16 + 48 + 36 = 104$

c How many of them are divisible by 5?

Must end in 5

1-digit: 1

2-digit: 4

3-digit: $4 \times 3 = 12$

4-digit (start 1 or 3, end 5):

Start 1: $3 \times 2 = 6$

Start 3: $3 \times 2 = 6$

Total: $1 + 4 + 12 + 12 = 29$

d How many of them are divisible by 3?

Sum of digits must be divisible by 3.

1-digit possibilities: 3, 9: **2**

2-digit digit possibilities: (1,5), (1,8), (3,9); 3 ways, multiply by 2! ways to rearrange each set: **6**

3-digit possibilities: {1,3,5}, {1,3,8}, {1,5,9}, {1,8,9}, 4 ways, multiply by 3! ways to rearrange each set: **24**

4-digit starting with 1: {1, 3, 8, 9}, {1, 3, 5, 9}, 2 ways, multiply by 3! ways to rearrange each set (since 1 must remain fixed at start): **12**

4-digit starting with 3: (3, 1, 8, 9}, {3, 1, 5, 9}, multiply by 3! ways to rearrange each set (since 1 must remain fixed at start): **12**

Total: $2 + 6 + 24 + 12 + 12 = 56$

Permutations with Identical Elements

Investigation Ordered selection with identical elements.

Consider the word **ALLY**.
To understand why we need to account for identical letters,
we'll temporarily label the two Ls as L_1 and L_2.
We can arrange 4 different letters in 4P4 = 4! = 24 ways.

A L_1 L_2 Y	L_1 A L_2 Y	L_2 A L_1 Y	Y A L_1 L_2
A L_2 L_1 Y	L_1 A Y L_2	L_2 A Y L_1	Y A L_2 L_1
A L_1 Y L_2	L_1 L_2 A Y	L_2 L_1 A Y	Y L_1 A L_2
A L_2 Y L_1	L_1 L_2 Y A	L_2 L_1 Y A	Y L_2 A L_1
A Y L_1 L_2	L_1 Y A L_2	L_2 Y A L_1	Y L_1 L_2 A
A Y L_2 L_1	L_1 Y L_2 A	L_2 Y L_1 A	Y L_2 L_1 A

Remove the labels and examine the list again. Notice that many arrangements are identical.

A L L Y	L A L Y	L A L Y	Y A L L
A L L Y	L A Y L	L A Y L	Y A L L
A L Y L	L L A Y	L L A Y	Y L A L
A L Y L	L L Y A	L L Y A	Y L A L
A Y L L	L Y A L	L Y A L	Y L L A
A Y L L	L Y L A	L Y L A	Y L L A

Each unique arrangement appears …… times because we can swap L_1 and L_2 without creating a new word.

Therefore, we have overcounted by a factor of ……, and the total ways to arrange ALLY is

……

Consider the word ALLLY. You might think we simply divide by 3 to account for three identical letters. Below are the arrangements of the three Ls in just the pattern A L L L Y (we won't list all 5! = 120 possible permutations):

A L_1 L_2 L_3 Y
A L_1 L_3 L_2 Y
A L_2 L_1 L_3 Y
A L_2 L_3 L_1 Y
A L_3 L_1 L_2 Y
A L_3 L_2 L_1 Y

How many ways can we rearrange the 3 Ls for each word we make? ……

Therefore, when we use 5!, we overcount by a factor of ……

The total number of ways to arrange the letters in the word ALLLY is:

……

ⓘ Permutations with Identical Elements

Number of different ways to arrange n objects where a objects are identical:

$$\frac{n!}{a!}$$

Number of different ways to arrange n objects where a objects are identical AND b different objects are identical:

$$\frac{n!}{a!\,b!}$$

Interpret Transformation

For each scenario, write an expression for the number of different arrangements.

a	TREE $\frac{4!}{2!}$	**b**	TAMMY	**c**	SYDNEY
d	CHEESE	**e**	BANANA	**f**	ADDRESSESS

g 3 identical wine glasses and 5 identical tumblers are to be arranged in a shelf.

h 4 red, 6 blue, 2 green marbles are to be arranged in a line. All the marbles are identical except for colour.

Example Count permutations with identical elements by considering cases.

How many 6-letter words can be formed by using the 7 letters of PRESSES?

Case 1: Leave out P
Letters: R, E, E, S, S, S

$$\frac{6!}{2! \times 3!} = \frac{720}{12} = 60$$

Case 2: Leave out R
Letters: P, E, E, S, S, S

$$\frac{6!}{2! \times 3!} = \frac{720}{12} = 60$$

Case 3: Leave out one E
Letters: P, R, E, S, S, S

$$\frac{6!}{3!} = \frac{720}{6} = 120$$

Case 4: Leave out one S
Letters: P, R, E, E, S, S

$$\frac{6!}{2! \times 2!} = \frac{720}{4} = 180$$

Total = 60 + 60 + 120 + 180 = 420

Practice

How many 5-letter words can be formed using the letters of PEPPER?

Case 1: Leave out P
Remaining: P, P, E, E, R
$\frac{5!}{2!\,2!} = 30$

Case 2: Leave out E
Remaining: P, P, P, E, R
$\frac{5!}{3!} = 20$

Case 3: Leave out R
Remaining: P, P, P, E, E
$\frac{5!}{3!\,2!} = 10$

Total = 30 + 20 + 10 = 60

In how many ways can the letters of MISSISSIPPI be arranged such that no two S's are adjacent?

Hint: use the gaps method

Arrange non-S numbers so there are gaps
_ M _ I _ I _ I _ P _ P _ I _
$\frac{7!}{4!\,2!}$

Arrange the 4 S into 8 gaps
$\frac{8P4}{4!}$

Total: $\frac{7!}{4!\,2!} \times \frac{8P4}{4!}$
7350

1 Find the number of permutations of the following words if all the letters are used.

a BOB

$\frac{3!}{2!} = 3$

b ALAN

$\frac{4!}{2!} = 12$

c SNEEZE

$\frac{6!}{3!} = 120$

d TASMANIA

$\frac{8!}{3!} = 6720$

e BEGINNER

$\frac{8!}{2!\,2!} = 10\,080$

f FOOTBALLS

$\frac{9!}{2!\,2!} = 90\,720$

g EQUILATERAL

$\frac{11!}{2!\,2!\,2!} = 4\,989\,600$

h COMMITTEE

$\frac{9!}{2!\,2!\,2!} = 45360$

i WOOLLOOMOOLOO

$\frac{14!}{7!\,3!} = 25\,720$

2 The six digits 1, 1, 1, 2, 2, 3 are used to form a six-digit number.
How many numbers can be formed?

$\frac{6!}{3! \times 2!} = 60$

3 Six coins are lined up on a table. Find how many patterns are possible if there are:

a five tails and one head

$\frac{6!}{5! \times 1!} = 6$

b four heads and two tails

$\frac{6!}{4! \times 2!} = 15$

c three tails and three heads

$\frac{6!}{3! \times 3!} = 20$

4 Eight balls, identical except for colour, are arranged in a line. Find how many different arrangements are possible if:

a all balls are of a different colour

$8! = 40320$

b there are seven red balls and one white ball

$\frac{8!}{7! \times 1!} = 8$

c there are six red balls, one white ball and one black ball

$\frac{8!}{6!} = 56$

d there are three red balls, three white balls and two black balls

$\frac{8!}{3! \times 3! \times 2!} = 560$

5 Five identical green chairs and three identical red chairs are arranged in a row.
Find how many arrangements are possible:

a if there are no restrictions

$\frac{8!}{5! \times 3!} = 56$

b if there must be a green chair on both ends

Fix green at both ends.
Need to select 6 middle positions from 3 green 3 red

$\frac{6!}{3!\,3!} = 20$

6 A motorist travels through eight sets of traffic lights, each of which is red or green. He is forced to stop at three sets of lights.

a In how many ways could this happen?

Arrange 3 red, 5 green

$$\frac{8!}{3! \times 5!} = 56$$

b What other number of red lights would give an identical answer to part a?

By the symmetric property, 5 red and 3 green.

7 In how many ways can the letters of the word SOCKS be arranged in a line:

a without restriction

$$\frac{5!}{2!} = 60$$

b so that the two Ss are together

Treat SS as unit: 4 unique units

$$4! = 24$$

c so that the two Ss are separated by at least one other letter

Total - Ss together

60 - 24 = 36

d so that the K is somewhere to the left of the C

By symmetry,

$$\frac{60}{2} = 30$$

8 Find the number of arrangements of the letters in SLOOPS if:

a there are no restrictions

$$\frac{6!}{2! \times 2!} = 180$$

b the two Os are together

Treat OO as unit: (OO, S, S, L, P)

$$\frac{5!}{2!} = 60$$

c the two Os are to be separated

Total - Os together

180 - 60 = 120

d the Os are together and the Ss are together

Treat OO as unit, SS as unit: 4 unique units

4! = 24

9 In how many arrangements of the letters in TATTOO are the two Os separated?

Total:

$$\frac{6!}{2! \times 2!} = 180$$

Os together:

$$\frac{5!}{2!} = 60$$

Os separated: 180 - 60 = 40

10 In how many ways can the letters of the word DECISIONS be arranged:

a without restriction

$\frac{9!}{2! \times 2!} = 90720$

b so that the vowels and consonants alternate

Vowels E,I,I,O (4), Consonants D,C,S,S,N (5)
Pattern CVCVCVCVC
Arrange consonants: $\frac{5!}{2!}$
Arrange vowels: $\frac{4!}{2!}$
$\left(\frac{5!}{2!}\right) \times \left(\frac{4!}{2!}\right) = 720$

c so that the vowels come first followed by the consonants

VVVVCCCCC
The counting is identical to having them alternate.
$\left(\frac{5!}{2!}\right) \times \left(\frac{4!}{2!}\right) = 720$

d so that the N is somewhere to the right of the D

Divide total by 2
$\frac{90720}{2} = 45360$

11 In how many ways can the letters of the word PROPORTIONALITY be arranged so that the vowels and consonants still occupy the same places?

Vowels O,O,I,O,A,I
Consonants P,R,P,R,T,N,L,T,Y
$\frac{6!}{3! \times 2!} \times \frac{9!}{2! \times 2! \times 2!} = 2721600$

12 A form has ten questions in order, each of which requires the answer 'Yes' or 'No'. Find the number of ways the form can be filled in:

a without restriction

$2^{10} = 1024$

b if the first and last answers are 'Yes'

Fix first and last as Y, fill 8 middle positions
$2^8 = 256$

c if two are 'Yes' and eight are 'No'

$\frac{10!}{2! \times 8!} = 45$

d if five are 'Yes' and five are 'No'

$\frac{10!}{5! \times 5!} = 252$

e if more than seven answers are 'Yes'

8 Yes: $\frac{10!}{8! \times 2!} = 45$
9 Yes: $\frac{10!}{9! \times 1!} = 10$
10 Yes: $\frac{10!}{10!} = 1$
45 + 10 + 1 = 56

f if an odd number of answers are 'Yes'

$\frac{1024}{2} = 512$

g if exactly three answers are 'Yes', and they occur together

Treat YYY as a unit. The rest are 7 No's
$\frac{8!}{7!} = 8$

h if the first and last answers are 'Yes' and exactly four more are 'Yes'

Fix first and last as Y.
Rest are 4 Yes and 4 No
$\frac{8!}{4! \times 4!} = 70$

13 How many five-letter words can be formed using the letters of the word:

a STRESS

Case 1: leave out an S
$\frac{5!}{2!} = 60$
Case 2: leave out a T
$\frac{5!}{3!} = 20$
Case 3: Leave out a R
$\frac{5!}{3!} = 20$
Case 4: Leave out an E
$\frac{5!}{3!} = 20$

$60 + 20 + 20 + 20 = 120$

b BANANA

Case 1: Leave out B
$\frac{5!}{3!\,2!} = 10$
Case 2: Leave out A
$\frac{5!}{2!\,2!} = 30$
Case 3: Leave out N
$\frac{5!}{3!} = 20$

$10 + 30 + 20 = 60$

14 Find how many arrangements of the letters of the word TRANSITION are possible if:

a there are no restrictions

10 letters, I twice, T twice, N twice
$\frac{10!}{2! \times 2! \times 2!} = 453\,600$

b the Is are together

Treat II as unit: 9 objects, T twice, N twice
$\frac{9!}{2! \times 2!} = 90720$

c the Is are together, and so are the Ns, and so are the Ts

Treat II, NN, TT as units: 7 objects
$7! = 5040$

d the Ns occupy the end positions

8 middle positions: I twice, T twice
$\frac{8!}{2! \times 2!} = 10080$

e an N occupies the first but not the last position

Fix first position as N.
Place N anywhere other than the last position: 8 options
Need to fill 8 positions. I twice, T twice
$8 \times \frac{8!}{2!\,2!} = 80640$

f the letter N is not at either end

Approach: Total – First – Last – First and Last
From part e, N first and not last is 80 640
N last and not first would also be 80 640
From part d, first and last is 10 080

453600 – 80640 – 80640– 10080 = 282 240

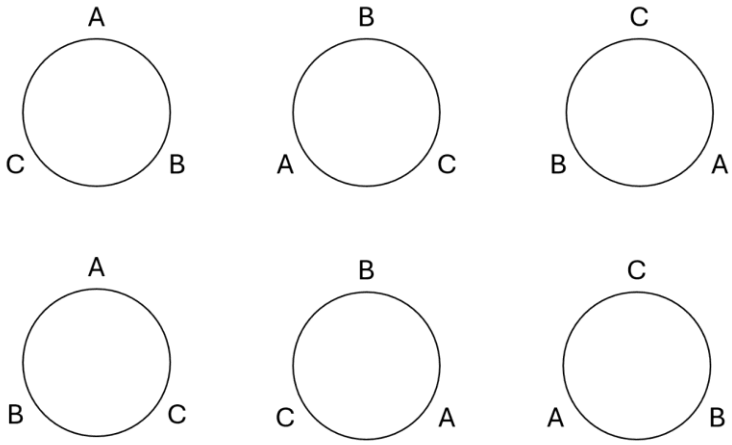
g the vowels are together

Treat (IIAO) as single object, so 7 objects (IIAO), T twice, N Twice
Arrange the units:
$\frac{7!}{2!\,2!}$
Arrange within unit:
$\frac{4!}{2!}$
Total:
$\frac{7!}{2!\,2!} \times \frac{4!}{2!} = 15\,120$

Arrangements around a Circle

Investigation Counting arrangements around a circle.

Consider a very small dinner party where you arrange three people, Ariel, Bec, and Cate around a circular table.



At first glance, there are 3! = …… arrangements.

However, circular arrangements have no clear beginning or end, unlike straight lines.
We can rotate the table (or simply walk around) and get the same arrangement.
Since we can rotate each arrangement 3 times, we've overcounted by a factor of 3.

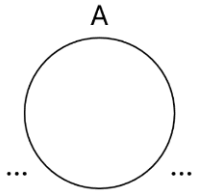
So, the total number of different arrangements are: ……

Another way to understand this is: only relative positions matter.

The first person sits anywhere (this counts as one way since we can rotate the table).

Then …… ways remain to arrange the other two people.

A	B	C
1	2	1



Arrangements around a Circle

Arranging n objects around a circle is the same as arranging $(n-1)$ objects in a line.

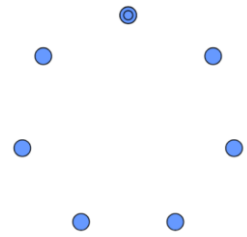
There is only one way to place the first object because all positions are initially equivalent due to rotation.

$$\frac{n!}{n} = (n-1)!$$

A diagram and the box method is useful for setting up these problems.

Example Count arrangements around a circle.

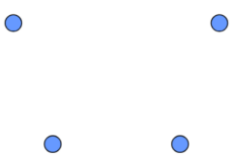
In how many ways can 7 people sit around a circular table?



1st	2nd	3rd	4th	5th	6th	7th
1	6	5	4	3	2	1

$$\frac{7!}{7} = 6! = 720$$

A, B, C, D, E are to be arranged around a circle. If B must be directly to the left of A, how many arrangements are there?

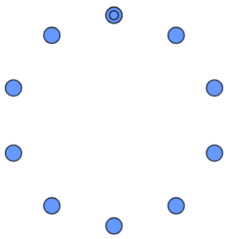


A	B	C	D	E
1	1	3	2	1

$$3! = 6$$

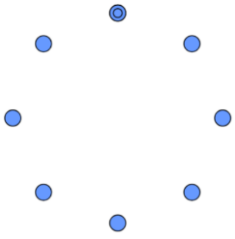
Practice

10 people sit around a round table. How many ways can this be done?



$9! = 24$

8 people sit around a round table. Bella and Zoe do not want to sit together. How many possible arrangements are there?



$1 \times 5 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 3600$

ⓘ Arrangements around a Circle involving Grouping

1. Identify the units.
2. Arrange the units around the circle.
3. Arrange the individuals within each unit.
4. Multiply the arrangements.

Example Count arrangements around a circle.

In how many ways can 6 people sit around a circular table, if two people must sit together?

1. Let units be (AB), C, D, E, F
2. Arrange 5 units around a circle: $4!$

1st	2nd	3rd	4th	5th
1	4	3	2	1

3. Arrangements of (AB): 2!
4. $4! \times 2! = 48$

6 people are to be arranged around a circular table. 3 friends wish to sit together as a group. How many ways can this arrangement occur?

1. (ABC), D, E, F
2. Order the 4 units around a circle: 3!

1st	2nd	3rd	4th
1	3	2	1

3. Arrangements of (ABC): 3!
4. $3! \times 3! = 36$

Practice

Five boys and five girls sit around a round table. There are two couples who wish to sit together. How many ways can this be done?

(BG), (BG), B, B, B, G, G, G
$7! \times 2! \times 2! = 20\,160$

Five boys and five girls sit around a round table. There are four couples who wish to sit together, and an ex-couple Shane and Penny who do not wish to sit together. How many ways can this be done?

(BG), (BG), (BG), (BG), S, P
Seat Shane first, then Penny, then 4 couples: $1 \times 3 \times 4 \times 3 \times 2 \times 1$
then rearrange within couples: $2! \times 2! \times 2! \times 2$
Total = 1152

1 A group of friends go into a restaurant and are seated around a circular table. Find how many arrangements are possible if there are

a 4 friends

$(4-1)! = 3! = 6$

b 7 friends

$(7-1)! = 6! = 720$

c 8 friends

$(8-1)! = 7! = 5040$

2 In how many ways can a group of 6 people be arranged

a in a line?

$6! = 720$

b in a circle?

$(6-1)! = 5! = 120$

3 Find how many different ways a group of 9 people can be arranged in

a in a line?

$9! = 362880$

b in a circle?

$(9-1)! = 8! = 40320$

4 Eight people are arranged so that two particular people sit together.
How many ways if the people are arranged in:

a in a line?

Treat two people as one unit: $7! = 5040$
Multiply by 2 (internal arrangements): $7!(2) = 10080$

b in a circle?

Treat two people as one unit in circle: $(7-1)! = 6! = 720$
Multiply by 2: $6!(2) = 1440$

5 Bob, Betty, Ben, Brad and Belinda are to be seated at a round table. In how many ways can this be done:

a if there are no restrictions,

$(5-1)! = 4! = 24$

b if Betty sits directly at Bob's right-hand side,

Bob	Betty	The rest
1	1	3!

Fix Bob's position, Betty must be on his right:
$3! = 6$

c if Brad is to sit directly between Bob and Ben,

Fix Bob's position, Ben can be in two positions, and Brad must be between (1 position)

Bob	Ben	Brad	The rest
1	2	1	2!

$2 \times 2! = 4$

$2(3-1)! = 2(2!) = 4$

d if Belinda and Betty sit apart,

Betty and Belinda together
Arrange 4 units around the table, then 2! internal arrangements
$(4-1)!(2!) = 12$

Total - (Belinda and Betty together)
$24 - 12 = 12$

e if Ben and Belinda sit apart, but Betty sits next to Bob?

Treat Betty and Bob as a single unit. 2! ways for internal arrangement.
Fix their position. Brad can only sit in one spot for Ben and Belinda to sit apart.
2 ways to arrange Ben and Belinda in the remaining two spots.
$2 \times 2! = 4$

6 Four boys and four girls are arranged in a circle. In how many ways can this be done:

a if there are no restrictions,

$(8-1)! = 7! = 5040$

b if the boys and the girls alternate,

Fix one boy's position, arrange 3 other boys: 3! = 6
Arrange 4 girls in gaps: 4! = 24
Total: 6(24) = 144

c if the boys and girls are in distinct groups,

Two units BBBBGGGG
Arrange 2 units around a circle: (2–1)!
Arrange boys 4!
Arrange girls 4!
$4! \times 4! = 576$

d if a particular boy and girl wish to sit next to one another,

Treat them as 1 unit. There are now 7 units.
Arrange 7 units around a circle (7–1)!
Internal arrangement of boy and girl 2!
$6! \times 2! = 1440$

e if two particular boys do not wish to sit next to one another,

Total – 2 particular people sitting together
$5040 - 1440 = 3600$

f if one particular boy wants to sit between two particular girls.

Treat GBG as 1 group. Internal rearrangement of girls 2!
Arrange 6 units around a circle (6–1)!
$2! \times 5! = 240$

7 Consider a necklace of six differently coloured beads. Because the necklace can be turned over, clockwise and anti-clockwise arrangements of the beads do not yield different orders. Hence find how many different arrangements there are of the six beads on the necklace.

$\frac{(6-1)!}{2} = 60$

8 In how many ways can ten different keys be placed on a key ring?

$\frac{(10-1)!}{2} = 181\,440$

9 In how many ways can a set of 10 beads be arranged

a in a line

$10! = 3628800$

b in a circle?

$(10-1)! = 9! = 362880$

c on a bracelet?

$\frac{(10-1)!}{2} = \frac{9!}{2} = 181440$

10 A group of 7 people sit around a table. In how many ways can they be arranged

a with no restrictions

$(7-1)! = 6! = 720$

b if 2 people want to sit together

Treat pair as one unit, 2! internal arrangements
arrange 6 units around a circle: 5!
$2!\,5! = 240$

c if 2 people cannot sit together

Total - (2 together): 720 - 240 = 480

d if 3 people sit together?

Treat three as one unit, 3! internal arrangements
arrange 5 units around a circle: 4!
$3!\,4! = 144$

11 In how many ways can the integers 1, 2, 3, 4, 5, 6, 7, 8 be placed in a circle if:

a there are no restrictions,

$(8-1)! = 7! = 5040$

b all the even numbers are together,

Even numbers (2,4,6,8): 4 numbers
Treat as one unit with odd numbers (1,3,5,7): 5 objects
Circular: (5-1)! = 4!
Internal even arrangements: 4!
Total: 4! 4! = 576

c the odd and even numbers alternate,

Fix one odd number, arrange 3 others: 3!
Arrange 4 even in gaps: 4!
Total: 3! 4! = 144

d at least two odd numbers are together,

Approach: Total - (no odd together)
Consider alternating (no 3 odd together): from part c = 144

5040 - 144 = 4896

e the numbers 1 and 7 are adjacent,

Treat 1 and 7 as one unit: (7-1)! = 6! = 720
Multiply by 2: 720(2) = 1440

f the numbers 3 and 4 are separated?

total - (3 and 4 adjacent): 5040 - 1440 = 3600

12 Find how many arrangements of n people around a circle are possible if:

a there are no restrictions,

$(n-1)!$

b two particular people must sit together,

Treat pair as unit, there are $n-1$ units.
Arrange around circle: = $(n-2)!$
Multiply by 2! Internal arrangements: 2! $(n-2)!$

c two particular people sit apart,

Total - together: $(n-1)! - 2(n-2)!$
$= (n-2)!\,(n-1-2)$
$= (n-2)!\,(n-3)$

d three particular people sit together.

Treat 3 as unit, there are $n-2$ units.
Arrange around circle = $(n-3)!$
Multiply by 3!: $6(n-3)!$

13 Twelve marbles are to be placed in a circle. In how many ways can this be done if:

a all the marbles are of different colours,

$(12-1)! = 11! = 39\,916\,800$

b there are eight red, three blue and one green marble?

Identical items (8 red, 3 blue, 1 green)
Fix one position, then divide to account for identical elements
$\frac{11!}{8!3!1!} = 165$

14 A sports committee consisting of four rowers, three basketballers and two cricketers sit at a circular table.

a How many different arrangements of the committee are possible if the rowers and basketballers both sit together in groups, but no rower sits next to a basketballer?

Treat as one big group: {RRRR C BBB},
Choose 1 of 2 cricketers to sit in middle: 2
The problem has been simplified to the 1 big unit and a lone cricketer, 2 units total, arrange 2 units around circle: 1!
Arrange rowers: 4!
Arrange basketballers: 3!
Total: $2 \times 4! \times 3! = 288$

b One rower and one cricketer are related. If the conditions in part **a** apply, what is the probability that these two members of the committee will sit next to one another?

The cricketers and rowers are seated in this manner: CRRRRC, there is a 1 in 4 chance the rower sits next to the cricketer he/she's related to

Unordered Selections without Repetition (Combinations)

Investigation Unordered selections without repetition.

Permutations involve **ordered** arrangements from a larger set.

Examples include: Arranging in a line, choosing teams where positions matter.

Combinations involve choosing **unordered** subsets from a larger set.

Examples include: Choosing teams where positions don't matter.

You have 5 people: A, B, C, D, E. You need to form a team of 3 where order doesn't matter.

Method 1: List systematically (there are 10)

Method 2: Divide permutations by identical subsets.

The number of ordered arrangements of 3 people from 5 is: ……

Consider the team of {A, B, C}. Since order doesn't matter:

A, B, C A, C, B B, A, C B, C, A C, A, B C, B, A are all the same team.

Each team can be arranged in ……! = …… ways, so we divide the total permutations by ……!

Therefore, total number of combinations is: $\frac{(\ )P(\ )}{(\ )!}$ which can be simplified to:

Method 3: Yes-or-No method

Think of asking each person: "Are you in the team?"

For example, choosing {A, C, D} from A B C D E is the same as Y N Y Y N.

Order matters here as Y Y Y N N {A, B, C} is different to {A, C, D}.

For a team of 3, this is equivalent to arranging 3 Ys and 2 Ns in a line, which is:

ⓘ Combinations

A **combination** is an **unordered** selection of distinct elements from a set without repetition.

To select r objects from a set of n objects:

$$^nC_r = \frac{n!}{r!\,(n-r)!}$$

Interpret Combination notation.

Write an expression for:

a $^6C_2$

$= \frac{6!}{2!\,4!}$

b $^8C_2$

c $^8P_5$

d $^{10}C_{10}$

e $^{10}C_0$

f $^{11}C_1$

g $^{11}C_{10}$

h $^{10}C_2$

i $^{10}C_8$

What do you notice about ^nC_n and $^nC_0$?

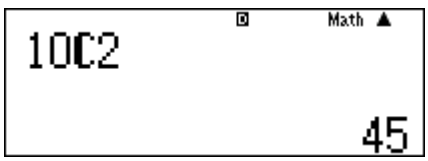
What do you notice about $^nC_1$ and $^nC_{n-1}$?

What do you notice about ^nC_r and $^nC_{n-r}$?

ⓘ Combination Notation on a Calculator

FX-82 AU

SHIFT ÷



FX-8200 AU

⊛ – [Probability] > [Combination(C)]

Example Count total number of combination arrangements.

A badminton team of 2 is chosen from 8 players. How many ways can this be done?

$${}^{8}C_{2} = 28$$

A school has 10 Physics teachers and 15 Maths teachers. A committee to is to be formed with 4 Physics teachers and 6 Maths teachers. How many ways can this be done?

$${}^{10}C_{4} \times {}^{15}C_{6} = 1\,051\,050$$

Practice

10 people meet to play doubles tennis. How many ways can 4 people be selected to play the first game (ignore subsequent arrangement into pairs)

$10C4 = 210$

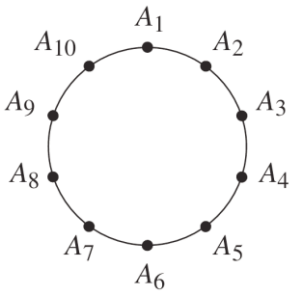
A coin is tossed 5 times. How many ways are there of getting 2 heads?

$5C2 = 10$

A workplace has 4 women and 6 men. How many ways can you select 2 women and 2 men to form a committee?

$4C2 \times 6C2 = 90$

Ten points are arranged equally around a circle. How many ways are there of drawing a triangle using these points?



$10C3 = 120$

Example Count total number of combination arrangements.

Let $S = \{2, 4, 6, 8, 10, 12\}$
How many subsets of S contain at most two elements?

Case 1: Sets with 0 numbers

${}^{6}C_{0} = 1$

Case 2: Sets with 1 number

${}^{6}C_{1} = 6$

Case 3: sets with 2 numbers

${}^{6}C_{2} = 15$

Ten cards are selected from a standard deck of 52. Find the number of combinations if exactly 7 of the cards are hearts.

Choose 7 hearts from the 13 hearts, then choose 3 other cards from remaining 39.

$${}^{13}C_{7} \times {}^{39}C_{3} = 15\,682\,524$$

Practice

10 students, 4 boys and 6 girls, travel from school to a sports ground. 6 go on a minibus, 2 take the train and 2 get driven by their parents. How many ways is this possible if none of the girls take the train and 1 boy and 1 girl get driven?

First deal with the restrictions:
4C2 boys take the train. 2 Boys 6 girls remain
$2C1 \times 6C1$ get driven. 1 boy 5 girls remain
$6C6$ take the minibus

$4C2 \times 2C1 \times 6C1 \times 6C6 = 72$ possible ways

A debating team of 5 members is to be chosen from 4 boys and 3 girls. How many ways are there of choosing the members so that boys outnumber girls?

Case 1: BBBGG: $4C3 \times 3C2 = 12$
Case 2: BBBBG $4C4 \times 3C1 = 3$
Case 3: BBBBB is impossible
Total: 15

1 Find in how many ways you can form a group of:

a two people from a choice of seven,

$^7C_2 = 21$

b three people from a choice of seven,

$^7C_3 = 35$

c two people from a choice of six,

$^6C_2 = 15$

d five people from a choice of nine.

$^9C_5 = 126$

2 A bag contains 23 lollies. If I take 6 lollies out, how many different combinations are possible?

$^{23}C_6 = 100\,947$

3 A team of 4 players is chosen at random from a group of 20 tennis players to play in an exhibition match. In how many ways could the team be chosen?

$^{20}C_4 = 4845$

4

a Find how many possible combinations there are if, from a group of ten people:

i two people are chosen,

$^{10}C_2 = 45$

ii eight people are chosen.

$^{10}C_8 = 45$

b Why are the answers identical?

Choosing 2 from 10 is the same as choosing 8 to leave behind.

This is the identity $^nC_r = {}^nC_{n-r}$

5 From a party of 12 men and 8 women, find how many groups can be chosen consisting of:

a five men and three women

$^{12}C_5 \times {}^8C_3 = 44\,352$

b four women and four men

$^8C_4 \times {}^{12}C_4 = 34650$

6 Four numbers are to be selected from the set of the first eight positive integers.
Find how many possible combinations there are if:

a there are no restrictions,

$^8C_4 = 70$

b there are two odd numbers and two even numbers,

Choose 2 odd from 4, then 2 even from 4
$^4C_2 \times {}^4C_2 = 36$

c there is exactly one odd number,

Choose 1 odd, then choose the rest from the even.
$4C1 \times 4C3 = 16$

d all the numbers must be even,

$^4C_4 = 1$

e there is at least one odd number.

Total – all even = 70 – 1 = 69

7 A bag contains 3 green and 6 blue balls. How many ways are there of drawing 4 balls if:

a the balls may be of any colour,

$^9C_4 = 126$

b there are exactly two green balls,

Choose 2 green and 2 blue
$^3C_2 \times {}^6C_2 = 45$

c there are at least two green balls,

At least two means two cases: 2 Green + 3 Green
2 green: 45 (from part b)
3 green: $^3C_3 \times {}^6C_1 = 6$
$45 + 6 = 51$

d there are more blue balls than green balls.

two cases: 3 blue, 4 blue
$^6C_3 \times {}^3C_1 + {}^6C_4 = 75$

8 A committee of 6 people is to be selected at random from a group of 11 men and 12 women.
Find the number of possible committees if:

a there is no restriction on who is on the committee

$^{23}C_6 = 100\,947$

b all committee members are to be male

$^{11}C_6 = 462$

c all committee members are to be female

$^{12}C_6 = 924$

d there are to be 3 men and 3 women

$^{11}C_3 \times {}^{12}C_3 = 36\,300$

e Anna is included

Fix Anna in the group. Choose 5 people from 22 remaining
$^{22}C_5 = 26\,334$

f Bruce is not included

Exclude Bruce. Choose 6 people from 22 remaining
$^{22}C_6 = 74\,613$

g there are to be 4 women and 2 men.

$^{12}C_6 \times {}^{11}C_2 = 27\,225$

9 Ten cards are selected at random from a set of 52 playing cards.
Find the number of combinations selected if:

a there are no restrictions
(answer in scientific notation correct to 3 significant figures)

$^{52}C_{10} = 1.58 \times 10^{10}$

b they are all hearts

Choose 10 hearts from 13 hearts
$^{13}C_{10} = 286$

c there are 7 hearts

Choose 7 hearts from 13, then 3 from the rest
$^{13}C_7 \times {}^{39}C_3 = 15\,682\,524$

d they are all red cards

Choose 10 red from 26 total
$^{26}C_{10} = 5\,311\,735$

e there are 4 aces.

Choose 4 aces from 4, then 6 from 48 remaining cards
$^{4}C_4 \times {}^{48}C_6 = 12\,271\,512$

10 Out of a group of 25 students, 7 walk to school, 12 catch a train and 6 catch a bus.
If 6 students are selected, find the number of combinations if:

a all walk to school

$^{7}C_6 = 7$

b no one catches a bus

Choose 6 from 19 who walk/train
$^{19}C_6 = 27132$

c 3 walk to school and 1 catches a bus

$^{7}C_3 \times {}^{6}C_1 \times {}^{12}C_2 = 13860$

d 1 walks to school and 4 catch a train

$^{7}C_1 \times {}^{12}C_4 \times {}^{6}C_1 = 20790$

e 3 catch a train and 1 catches a bus.

$^{12}C_3 \times {}^{6}C_1 \times {}^{7}C_2 = 27720$

11 A committee of five is to be chosen from six men and eight women.
Find how many committees are possible if:

a there are no restrictions,

$^{14}C_5 = 2002$

b all members are to be female,

$^{8}C_5 = 56$

c all members are to be male,

$^{6}C_5 = 6$

d there are exactly two men,

$^{6}C_2 \times {}^{8}C_3 = 840$

e there are four women and one man,

$^{8}C_4 \times {}^{6}C_1 = 420$

f there is a majority of women,

three cases: 3 women, 4 women, or 5 women
$(^{8}C_3 \times {}^{6}C_2) + (^{8}C_4 \times {}^{6}C_1) + {}^{8}C_5 = 1316$

g a particular man must be included,

$^{13}C_4 = 715$

h a particular man must not be included.

$^{13}C_5 = 1287$

12

a What is the number of combinations of the letters of the word EQUATION taken four at a time (without repetition)?

$^8C_4 = 70$

b How many of the four-letter combinations contain four vowels?

$^5C_4 = 5$

c How many of the four-letter combinations contain the letter Q?

$^7C_3 = 35$

13 A team of seven netballers is to be chosen from a squad of twelve players A, B, C, D, E, F, G, H, I, J, K and L. In how many ways can they be chosen:

a with no restrictions,

$^{12}C_7 = 792$

b if the captain C is to be included,

$11C6 = 462$

c if J and K are both to be excluded,

$^{10}C_7 = 120$

d if A is included but H is not,

$^{10}C_6 = 210$

e if one of F and L is to be included and the other excluded?

2 ways to choose between F and L, then fill 6 positions from 10 remaining
$^{10}C_6 \times 2 = 420$

14 Consider the digits 9, 8, 7, 6, 5, 4, 3, 2, 1 and 0.
Find how many five-digit numbers are possible if the digits are to be in:

a descending order,

Choose 5 from 10, arrange in descending order
$^{10}C_5 = 252$

b ascending order.

Choose 5 from 10, but numbers cannot start with 0; subtract cases where 0 is selected
$^{10}C_5 - {}^9C_4 = 252 - 126 = 126$

c Why do these two questions involve unordered selections?

Once 5 numbers selected, only one arrangement for ascending/descending is possible, so we do not need to count rearrangements.

15 Twelve people arrive at a restaurant. There is one table for six, one table for four and one table for two. In how many ways can they be assigned to a table?

Choose 6 people for the first table, then 4 for the next, and 2 for the last.
$^{12}C_6 \times {}^6C_4 \times {}^2C_2 = 13860$

16 Twenty students, ten male and ten female, are to travel from school to the sports ground. Eight of them go in a minibus, six of them in cars, four of them on bikes and two walk.

a In how many ways can they be distributed for the trip?

$^{20}C_8 \times {}^{12}C_6 \times {}^{6}C_4 \times {}^{2}C_2 = 1\,745\,944\,200$

b In how many ways can they be distributed if none of the boys walk?

$^{10}C_2 \times {}^{18}C_8 \times {}^{10}C_6 \times {}^{4}C_4 = 413\,513\,100$

17 Ten points P_1, P_2, ..., P_{10} are chosen in a plane, no three of the points being collinear.

a How many lines can be drawn through pairs of the points?

Choose 2 points to draw a line through
$^{10}C_2 = 45$

b How many triangles can be drawn using the given points as vertices?

Choose three points to connect as a triangle
$^{10}C_3 = 120$

c How many of these triangles have P_1 as one of their vertices?

Fix P1 and choose 2 points from 9 remaining
$^{9}C_2 = 36$

d How many of these triangles have P_1 and P_2 as vertices?

Fix P1 and P2, choose 1 more point from 8 to make a triangle
$^{8}C_1 = 8$

18 Ten points are chosen in a plane. Five of the points are collinear, but no other set of three of the points is collinear.

a How many sets of three points can be selected from the five that are collinear?

Choose 3 points from the 5 collinear
$^{5}C_3 = 10$

b How many triangles can be formed using three of the ten points as vertices?

Total ways to choose 3 points – collinear ways
$^{10}C_3 - {}^{5}C_3 = 110$

19 From a standard deck of 52 playing cards, find how many five-card hands can be dealt:

a consisting of black cards only,

$^{26}C_5 = 65\,780$

b consisting of diamonds only,

$13C5 = 1287$

c containing all four kings,

$^{4}C_4 \times {}^{48}C_1 = 48$

d consisting of three diamonds and two clubs,

$^{13}C_3 \times {}^{13}C_2 = 22\,308$

e consisting of three twos and another pair,

4C3 ways to choose three twos
Then choose any rank from the 12 remaining
Then choose 2 cards from that rank
$^{4}C_3 \times {}^{12}C_1 \times {}^{4}C_2 = 288$

f consisting of one pair and three of a kind.

For the pair: Choose 1 rank from 13, then 2 from that rank
For the three of a kind: Choose 1 rank from 12 (since you cant have a pair and a 3 from the same rank) then choose 3 from that rank.
$(^{13}C_1 \times {}^{4}C_2) \times (^{12}C_1 \times {}^{4}C_3) = 3744$

20 A group of four people A, B, C and D are to be split into two groups of two.
Jimmy claims that the answer is ${}^4C_2 \times {}^2C_2 = 6$: first choose two people out of four to go into one group, then the remaining two people become the other group, by default.

a By listing the possible combinations, how many possible ways can the four people be split into two groups of two?

{A,B} & {C,D} {A,C} & {B,D} {A,D} & {B,C}

Only 3

b Jimmy insists it's 6, showing you his list:
Explain why his answer is exactly double the actual number of possible combinations.

Group 1	Group 2
A, B	C, D
A, C	B, D
A, D	B, C
C, D	A, B
B, D	A, C
B, C	A, D

Since group 1 and group 2 are indistinguishable (no Group 1 vs Group 2 labels) Bob's method overcounts by a factor of 2.

21 In how many ways can a group of six people be divided into:

a two unequal groups
(neither group being empty),

Possible unequal ways to split the group are (1, 5) and (2,3)
${}^6C_1 + {}^6C_2 = 21$

b two equal groups?

6C3, but we need to divide by 2 as the groups are indistinguishable
$\frac{{}^6C_3}{2} = 10$

c Repeat part **a** for four people

${}^4C_1 = 4$

d Repeat part **b** for four people

$\frac{{}^4C_2}{2} = 3$

e Repeat part **a** for eight people

${}^8C_1 + {}^8C_2 + {}^8C_3 = 92$

f Repeat part **b** for eight people

$\frac{{}^8C_4}{2} = 35$

22 Find how many diagonals there are in:

a a quadrilateral,

A quadrilateral has 4 vertices. Choose 2 points to connect from 4.
However, 4 of these lines are sides (a quadrilateral has 4 sides)
${}^4C_2 - 4 = 2$

Alternative method:
Consider each vertex, there are 3 other vertices you could connect to,
but 2 of those connections are sides.
Only 1 of those connections is a diagonal
So, 1×4
But you have double counted as v_1v_2 is the same as v_2v_1
$4 \div 2 = 2$

b a pentagon,

A pentagon has 5 vertices. Choose 2 points to connect from 5.
However, 5 of these lines are sides
${}^5C_2 - 5 = 5$

Alternative method:
Consider each vertex, there are 4 other vertices you could connect to,
but 2 of those connections are sides.
Only 2 of those connections is a diagonal
So $2 \times 5 = 10$
But you have double counted as v_1v_2 is the same as v_2v_1
$10 \div 2 = 5$

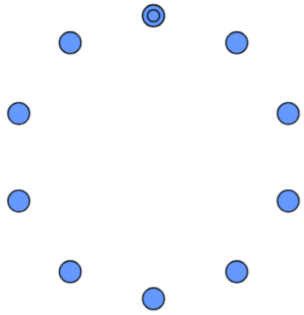
c a decagon,

${}^{10}C_2 - 10 = 35$

d a polygon with n sides.

${}^nC_2 - n$
or $\frac{(n-3) \times n}{2}$
if you consider both methods,
you have a combinatorial argument for this identity.

23 Ten points A, B, ..., J are arranged in order around a circle.



a How many triangles can be drawn with these points as vertices?

${}^{10}C_3 = 120$

b How many pairs of such triangles can be drawn, if the vertices of the two triangles are distinct?

Choose 6 distinct points to act as vertices. ${}^{10}C_6$

Choose 3 points for the first triangle, once you do this you have automatically chosen 3 for the other. ${}^{6}C_3$

However, since the two triangles are indistinguishable, divide by 2. $\frac{{}^{6}C_3}{2}$

E.g. Triangle 1: {A B C} Triangle 2: {D E F} is the same as Triangle 1: {D E F} Triangle 2: {A B C}

$${}^{10}C_6 \times \frac{{}^{6}C_3}{2} = 2100$$

c In how many such pairs will the triangles:

i not overlap,

Hint: For the triangles to not overlap, the points must be in consecutive order. List the ways.

For triangles to not overlap, we must choose consecutive points for each triangle.

Imagine you chose {A B C D E F} as your points, there are exactly three ways to partition them into two groups in consecutive order.

$\{A\ B\ C\}\{D\ E\ F\}$

$\{B\ C\ D\}\{E\ F\ A\}$

$\{C\ D\ E\}\{F\ A\ B\}$

${}^{10}C_6 \times 3 = 630$

ii overlap?

Not overlapping and overlapping are complementary

2100 - 630 = 1470

24 Twelve points are arranged in order around a circle.

a How many triangles can be drawn with these points as vertices?

${}^{12}C_3 = 220$

b In how many pairs of such triangles are the vertices of the two triangles distinct?

$${}^{12}C_6 \times \frac{{}^{6}C_3}{2} = 9240$$

c In how many such pairs will the triangles not overlap?

Like the previous question, three ways to partition six elements into two groups in consecutive order

${}^{12}C_6 \times 3 = 2772$

d In how many such pairs will the triangles overlap?

$9240 - 2772 = 6468$

25 Let S = {1, 3, 5, 7, 9, 11, 13, 15, 17, 19} be the set of the first ten positive odd integers.

a How many subsets does S have?

$2^{10} = 1024$

b How many subsets of S contain at least three numbers?

Subtract subsets of 0 numbers, 1 number, 2 numbers from total.
$2^{10} - {}^{10}C_0 - {}^{10}C_1 - {}^{10}C_2 = 968$

c How many subsets with at least three numbers do not contain 7?

Consider 9 elements (exclude 7)
$2^9 - {}^9C_0 - {}^9C_1 - {}^9C_2 = 466$

d How many subsets with at least three numbers do not contain 7 but do contain 19?

Fix 19, at least 2 more from remaining 8 (excluding 7)
${}^8C_2 + {}^8C_3 + {}^8C_4 + {}^8C_5 + {}^8C_6 + {}^8C_7 + {}^8C_8 = 247$

26 In how many ways can two numbers be selected from 1, 2, ..., 8, 9 so that their sum is:

a even,

There are 5 odd and 4 even.
A sum is even if both odd or both even
${}^5C_2 + {}^4C_2 = 16$

b odd,

a sum is odd if one odd, one even
$5 \times 4 = 20$

c divisible by 3,

list possible ways to sum to a multiple of 3:
(1,2), (1,5), (1,8), (2,4), (2,7), (3,6), (3,9), (4,5), (4,8), (5,7), (6,9), (7,8)
12

d divisible by 5,

(1,4), (1,9), (2,3), (2,8), (3,7), (4,6), (5,9), (6,9)
8

e divisible by 6?

List possible ways to sum to a multiple of 6. Can use part **c:**
(1,5), (3,9), (4,8), (5,7), (6,9)
5

27 There are ten basketballers in a team. Find in how many ways:

a the starting five can be chosen,

${}^{10}C_5 = 252$

b they can be split into two teams of five.

$\frac{{}^{10}C_5}{2} = 126$

28 Nine players are to be divided into two teams of four and one umpire.

a In how many ways can the teams be formed?

Choose umpire, then divide remaining 8 into two teams
${}^9C_1 \times \frac{{}^8C_4}{2} = 315$

b If two particular people cannot be on the same team, how many different combinations are possible?

total – same team.

Choose umpire, fix 2 in same team.
${}^7C_1 \times {}^6C_2 \times {}^4C_4 = 105$

$315 - 105 = 210$

29 Consider the number 18, it has six factors: 1, 2, 3, 6, 9, 18.
We can count the number of factors by using the powers of the prime factorisation:

$$18 = 2 \cdot 3^2$$

Each factor can be made from choosing a power for each prime (up to its maximum power).

For the prime 2, you can choose powers of 0 or 1.

For the prime 3, you can choose powers of 0, 1, or 2.

Therefore, there are $2 \times 3 = 6$ total choices, which matches the six factors.

By considering their prime factorisations, find the number of factors of:

a $2^3 \times 3^2$

$(3+1)(2+1) = 12$

b 1 000 000

$10^6 = 2^6 \cdot 5^6$
$(6+1)(6+1) = 49$

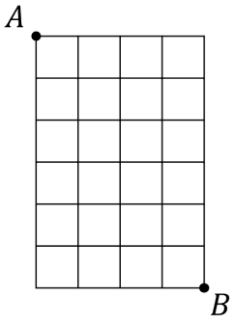
c 315 000

$315000 = 2^3 \cdot 3^2 \cdot 5^4 \cdot 7$
$(3+1)(2+1)(4+1)(1+1) = 120$

d $2^a \times 5^b \times 13^c$

$(a+1)(b+1)(c+1)$

30 The diagram shows a 6 × 4 grid. The aim is to walk from point A to the point B by walking along the black lines either downwards or to the right.
A single move is defined as walking along one side of a single small square, thus it always takes you ten moves to get from A to B.



By treating paths as ordered sequences of D and R:

a Find how many different routes are possible:

i without restriction,

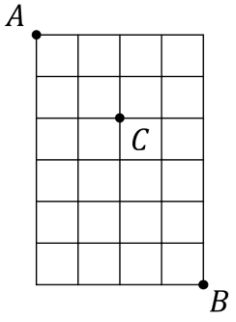
Need to count arrangements of 6 Ds and 4 Rs

This is the same as choosing 6 positions to get D, the rest automatically get R.

$\frac{10!}{6!\,4!} = {}^{10}C_6 = 210$

(Alternatively, choose 4 positions to get R, the rest automatically get D. ${}^{10}C_6 = {}^{10}C_4$ due to symmetric property.)

ii if you must pass through C,



C is 2 down, 2 right

Need 2 Ds 2 Rs at the start of the sequence, then 4 Ds 2 Rs

${}^4C_2 \times {}^6C_4 = 90$

iii if you cannot move along the top line of the grid,

First must be D

Arrange 5 Ds in 9 positions

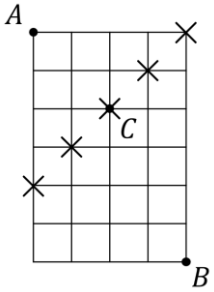
${}^9C_5 = 126$

iv if you cannot move along the second row from the top of the grid.

This is the same as part iii

${}^9C_5 = 126$

b Notice that every route must pass through one and only one of the five crossed points.
This is because a path must cross a diagonal to get from A to B.
Hence prove that ${}^{10}C_4 = {}^4C_0 \times {}^6C_4 + {}^4C_1 \times {}^6C_3 + {}^4C_2 \times {}^6C_2 + {}^4C_3 \times {}^6C_1 + {}^4C_4 \times {}^6C_0$.



Total number of paths, from part **a i**, is ${}^{10}C_6 = {}^{10}C_4$

Using the same method as part **a ii**

To pass through the leftmost crossed point you must have 4 Ds at the start, then 2 D 4 R = ${}^4C_0 \times {}^6C_4$

Similarly for the second crossed point from the left, you need 3 Ds 1 R at the start, then 3 Ds 3Rs = ${}^4C_1 \times {}^6C_2$

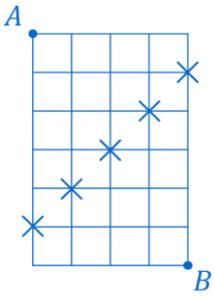
For the third crossed point = ${}^4C_2 \times {}^6C_1$

Fourth crossed point = ${}^4C_4 \times {}^6C_6$

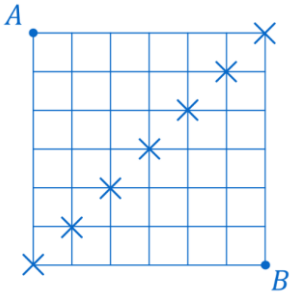
Since all the paths must pass through one of the five crossed points, we add up all the cases and it would equal total number of paths ${}^{10}C_4$

$\therefore {}^{10}C_4 = {}^4C_0 \times {}^6C_4 + {}^4C_1 \times {}^6C_3 + {}^4C_2 \times {}^6C_2 + {}^4C_3 \times {}^6C_1 + {}^4C_4 \times {}^6C_0$

c What diagonal would you draw to prove that
${}^{10}C_4 = {}^5C_0 \times {}^5C_4 + {}^5C_1 \times {}^5C_3 + {}^5C_2 \times {}^5C_2 + {}^5C_3 \times {}^5C_1 + {}^5C_4 \times {}^5C_0$.



d Using the 6 × 6 grid, then using the same idea as that used in parts b and c, prove that
${}^{12}C_6 = ({}^6C_0)^2 + ({}^6C_1)^2 + ({}^6C_2)^2 + ({}^6C_3)^2 + ({}^6C_4)^2 + ({}^6C_5)^2 + ({}^6C_6)^2$.



Combination Proofs and Identities

ⓘ Combination Identities

$^nC_n = {}^nC_0 = 1$

$^nC_1 = {}^nC_{n-1} = n$

$^nC_r = {}^nC_{n-r}$

$^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$

$^nC_0 + {}^nC_1 + {}^nC_2 \ldots {}^nC_n = 2^n$

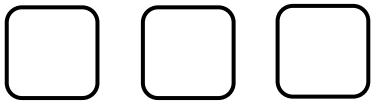
Proof $^nC_r = {}^nC_{n-r}$ (Symmetry Property)

Method 1: Algebra

LHS $=$ RHS $=$

Method 2: Combinatorial Argument





Suppose we are choosing 3 people from 5 for a team.
We can do this in two ways:

- Choose 3 people from 5 to be in the team: ……
- Choose … people from 5 to *not* be in the team: ……

Therefore, $^5C_3 = {}^5C_2$

Suppose we are choosing r people from n for a team.
We can do this in two ways:

- Choose …… people from … to be in the team: ……
- Choose …… people from … to *not* be in the team: ……

Therefore, $^nC_r = {}^nC_{n-r}$

Proof ${}^{n}C_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$ (Pascal's Identity, or Addition Property)

Another combinatorial identity is ${}^{n}C_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$

Method 1: Algebra

Hint: Make the common denominator $r!\,(n-r)!$

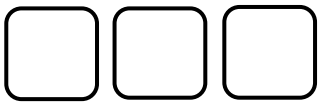
$\text{RHS} = {}^{n-1}C_{r-1} + {}^{n-1}C_r$

Method 2: Combinatorial Argument



Suppose we are choosing 3 people from 5 for a team.

In how many ways can you do this? ……



Let's consider a specific person, let's call her Alice.

- Case 1: Pick Alice to be on the team.
  You've filled 1 spot, so you need to choose …… people from the …… remaining: ……
- Case 2: Exclude Alice from the team.
  So, you need to choose …… people from the …… remaining: ……

  These two cases make up all possibilities. Therefore:

  ${}^{5}C_3 = {}^{4}C_2 + {}^{4}C_3$

Suppose we are choosing r people from n for a team.

In how many ways can you do this? ……

Let's consider a specific person, let's call them x.

- Case 1: Pick x to be on the team.
  You've filled 1 spot, so you need to choose …… people from the …… remaining: ……
- Case 2: Exclude x from the team.
  So, you need to choose …… people from the …… remaining: ……

  These two cases make up all possibilities. Therefore:

  ${}^{n}C_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$

Proof ${}^nC_0 + {}^nC_1 + {}^nC_2 \ldots {}^nC_n = 2^n$ (Binomial Identity)

Another combinatorial identity is ${}^nC_0 + {}^nC_1 + {}^nC_2 \ldots {}^nC_n = 2^n$

We cannot prove this using algebra (yet) until we learn proof by induction

Combinatorial Argument

Consider the set $S = \{a, b, c\}$ with 3 elements.
List all possible subsets of S, including the empty set and S itself.

0 element subsets 1 element subsets 2 element subsets 3 element subsets

How many subsets are there of each size? How does this relate to combinations ${}^nC_r$?

How many subsets did you find in total?

So, we can conclude that ${}^3C_0 + {}^3C_1 + {}^3C_2 + {}^3C_3 = \ldots\ldots$

So, the LHS can be thought of as counting ……
But how is this related to the RHS?

An alternate method of counting subsets is to decide whether to include each element.
This is a binary choice of yes or no:

Include a?	Include b?	Include c?

By the multiplication principle, the total number of ways is ……

Generalising:

For a set with n elements:

How many ways can you choose a subset with exactly r elements? ……

What is the total number of subsets of all possible sizes? ……

For a set with n elements, each element can either be included in a subset or excluded.

Using the multiplication principle, how many total subsets are there? ……

Therefore, we have proved ${}^nC_0 + {}^nC_1 + {}^nC_2 \ldots {}^nC_n = 2^n$

1 Solve for n.

a $^{10}C_n = {}^{10}C_3$

$n = 3$ or 7

b $^{n}C_7 = {}^{n}C_3$

$n = 10$

c $^{n}C_6 + {}^{n}C_5 = {}^{11}C_6$

Using Pascal's identity,
$n = 10$

d $^{8}C_n + {}^{8}C_{n-1} = {}^{9}C_3$

Using Pascal's identity
$n = 3$ or 6

2 Show that

$$^{2n}P_{n+1} = 2n \times {}^{2n-1}P_n$$

$$LHS = \frac{(2n)!}{(2n - n - 1)!} = \frac{(2n)!}{(n - 1)!}$$

$$RHS = \frac{n \times (2n - 1)!}{(2n - 1 - n)!} = \frac{2n \times (2n - 1)!}{(n - 1)!} = \frac{(2n)!}{(n-1)!}$$

$$= LHS$$

3 Show that

$$^{n}C_k = \frac{n - k + 1}{k} \; {}^{n}C_{k-1}$$

$$LHS = \frac{n!}{k!\,(n - k)!}$$

$$RHS = \left(\frac{n - k + 1}{k}\right) \times \frac{n!}{(k - 1)!\,(n - k + 1)!}$$

$$= \frac{(n - k + 1) \times n!}{k(k - 1)!\,(n - k + 1)!}$$

$$= \frac{n!}{k!\,(n - k)!}$$

We will save the combinatorial argument proofs for the end of the booklet 😊

Probability and Mixed Applications

ⓘ Probability Problems

$$\text{Probability} = \frac{\text{Number of acceptable outcomes}}{\text{Total possible outcomes}}$$

Interpret Permutation and Combination Probability Questions

For these questions, the answer is provided. Annotate to explain where it comes from.

a Jenny has 12 pairs of earrings, 3 necklaces, 8 rings, and 2 watches. If Karla were to guess the combination of jewellery, what is the chance she guesses correctly?

$$\frac{1}{12 \times 3 \times 8 \times 2}$$

b In a lottery, a machine contains 45 balls, each numbered 1 to 45. 4 balls are drawn and players must guess the correct order to win. Lisa has 3 different entries, what is the probability that she wins?

$$\frac{3}{45P4}$$

c Trinity chooses 13 songs at random from 20 songs in her playlist. What is the probability that her two favourite songs are part of the playlist?

$$\frac{18C11}{20C13}$$

d 7 people are seated around a circular table. What is the probability that a group of 3 particular people will be sat together?

$$\frac{3!\,(5-1)!}{(7-1)!}$$

e A coin is tossed 10 times. What is the probability of getting 5 heads, in any order?

$$\frac{10C5}{2^{10}}$$

f A team of 6 men and 5 women is chosen at random from a group of 8 men and 9 women. Find the probability that neither Kaya nor Greg are in the team.

$$\frac{7C6 \times 8C5}{8C6 \times 9C5}$$

1 A PIN has 4 numbers. If I forget my PIN I am allowed 3 tries to get it right.
Find the probability that I get it within the 3 tries.

$\frac{3}{10^4} = \frac{3}{10\,000}$

2 A restaurant offers 7 main courses and 4 desserts, as well as 3 different types of coffee.

a How many different combinations of main course, dessert and coffee are possible?

$7 \times 4 \times 3 = 84$

b Find the probability that I randomly pick the combination most often voted as favourite.

$\frac{1}{84}$

3 A password consists of 2 letters followed by 5 numbers.
What is the probability that Izak randomly guesses the correct password?

$\frac{1}{26^2 \times 10^5} = \frac{1}{67\,600\,000}$

4 A set of cards is numbered 1 to 100 and 2 cards are chosen at random.

a How many different arrangements of ordered pairs are possible?

$100 \times 99 = 9900$

b What is the probability that a particular ordered pair is chosen?

$\frac{1}{9900}$

5 Each of 10 cards has a letter written on it from A to J. If 3 cards are selected in order at random, find the probability that they spell out CAB.

$\frac{1}{10 \times 9 \times 8} = \frac{1}{720}$

6 A bookshelf is to hold 5 mathematics books, 8 novels and 7 cookbooks.

a In how many different ways could they be arranged?
(Leave your answer in factorial notation.)

$20!$

b If the books are grouped in categories, in how many ways can they be arranged?
(Answer in factorial notation.)

3 units. Arrange units, then arrange within units.

$3! \times 5! \times 8! \times 7!$

c If one book is chosen at random, find the probability that it is a cookbook.

$\frac{7}{20}$

7 There are 5 numbers: 3, 4, 5, 5 and 6.

a How many different arrangements can be made from the numbers?

$\frac{5!}{2!} = 60$

b How many arrangements form numbers greater than 40 000?

Consider 3 cases: First digit 4, 5 or 6

First digit 4: 4!/2! = 12

First digit 5: 4! = 24

First digit 6: 4!/2! = 12

Total = 12 + 24 + 12 = 48

c How many form numbers less than 50000?

First digit 3 or 4

First digit 3: 4!/2! = 12

First digit 4: 4!/2! = 12

Total = 12 + 24 = 36

d If an arrangement is made at random, find the probability that it is less than 40000.

Less than 40 000 = Total – Greater than 40 000

60 – 48 = 12

$\frac{12}{60} = \frac{1}{5}$

8 Find the probability that if 10 people sit around a table at random, 2 particular people will be seated together.

Total: 9!

2 particular people sit together, treat as 1 unit. Arrange 9 units around a circle and arrange within group: $8! \times 2$

$\frac{8! \times 2}{9!} = \frac{2}{9}$

9 A basketball team of 5 players is selected at random from a group of 12 PE students.

a In how many ways can the team be selected?

${}^{12}C_5 = 792$

b Find the probability that Erik is selected as one of the team members.

$\frac{{}^{11}C_4}{{}^{12}C_5} = \frac{5}{12}$

c Find the probability that Erik and Jens are both selected.

$\frac{{}^{10}C_3}{{}^{12}C_5} = \frac{5}{33}$

10 A committee of three is to be selected from the nine members in a club.

a How many different committees can be formed?

$^9C_3 = 84$

b If there are five men in the club, what is the probability that the selected committee consists entirely of males?

$\frac{^5C_3}{^9C_3} = \frac{5}{42}$

11 The integers from 1 to 10 inclusive are written on ten separate pieces of paper. Four pieces of paper are drawn at random. Find the probability that:

a the four numbers drawn are 1, 2, 3 and 6,

$\frac{1}{^{10}C_4} = \frac{1}{210}$

b the number 9 is one of the numbers drawn,

$\frac{^9C_3}{^{10}C_4} = \frac{2}{5}$

c the number 8 is not drawn,

$\frac{^9C_4}{^{10}C_4} = \frac{3}{5}$

d the number 7 is drawn but the number 1 is not.

$\frac{^8C_3}{^{10}C_4} = \frac{4}{15}$

12 A bag contains three red, seven yellow and five blue balls. If three balls are drawn from the bag simultaneously, find the probability that:

a all three balls are yellow,

$\frac{^7C_3}{^{15}C_3} = \frac{1}{13}$

b all the balls are of the same colour,

$\frac{^3C_3 + {}^7C_3 + {}^5C_3}{^{15}C_3} = \frac{46}{455}$

c there are two red balls and one blue ball,

$\frac{^3C_2 \times {}^5C_1}{^{15}C_3} = \frac{3}{91}$

d all the balls are of different colours.

$\frac{^3C_1 \times {}^7C_1 \times {}^5C_1}{^{15}C_3} = \frac{3}{13}$

13 A sports committee of five members is to be chosen from eight AFL footballers and seven soccer players. Find the probability that the committee will contain:

a only AFL footballers,

$\frac{^8C_5}{^{15}C_5} = \frac{8}{429}$

b only soccer players,

$\frac{^7C_5}{^{15}C_5} = \frac{1}{143}$

c three soccer players and two AFL footballers,

$\frac{^7C_3 \times {}^8C_2}{^{15}C_5} = \frac{140}{429}$

d at least one soccer player,

$1 - \frac{8}{429} = \frac{421}{429}$

e at most one soccer player,

$\frac{^8C_5 + {}^7C_1 \times {}^8C_4}{^{15}C_5} = \frac{2}{11}$

f Ian, a particular soccer player.

$\frac{^{14}C_4}{^{15}C_5} = \frac{1}{3}$

14 From a standard pack of 52 cards, three are selected at random. Find the probability that:

a they are the jack of spades, the two of clubs and the seven of diamonds,

$\frac{1}{{}^{52}C_3} = \frac{1}{22\,100}$

b all three are aces,

$\frac{{}^{4}C_3}{{}^{52}C_3} = \frac{1}{5525}$

c they are all diamonds,

$\frac{{}^{13}C_3}{{}^{52}C_3} = \frac{11}{850}$

d they are all of the same suit,

$\frac{4 \times {}^{13}C_3}{{}^{52}C_3} = \frac{22}{425}$

e they are all picture cards,

$\frac{{}^{12}C_3}{{}^{52}C_3} = \frac{11}{1105}$

f two are red and one is black,

$\frac{{}^{26}C_2 \times {}^{26}C_1}{{}^{52}C_3} = \frac{13}{34}$

g one is a seven, one is an eight and one is a nine,

$\frac{{}^{4}C_1 \times {}^{4}C_1 \times {}^{4}C_1}{{}^{52}C_3} = \frac{16}{5525}$

h two are sevens and one is a six,

$\frac{{}^{4}C_2 \times {}^{4}C_1}{{}^{52}C_3} = \frac{6}{5225}$

i exactly one is a diamond,

$\frac{{}^{13}C_1 \times {}^{39}C_2}{{}^{52}C_3} = \frac{741}{1700}$

j at least two of them are diamonds.

$\frac{{}^{13}C_2 \times {}^{39}C_1 + {}^{13}C_3}{{}^{52}C_3} = \frac{64}{425}$

15 Repeat the previous question if the cards are selected from the pack one at a time, and each card is replaced before the next one is drawn.

a they are the jack of spades, the two of clubs and the seven of diamonds,

$\left(\frac{1}{52}\right)^3$

b all three are aces,

$\left(\frac{4}{52}\right)^3$

c they are all diamonds,

$\left(\frac{13}{52}\right)^3$

d they are all of the same suit,

$4 \times \left(\frac{13}{52}\right)^3$

e they are all picture cards,

$\left(\frac{12}{52}\right)^3$

f two are red and one is black,

$3 \times \left(\frac{26}{52}\right)^2 \times \left(\frac{26}{52}\right)$

g one is a seven, one is an eight and one is a nine,

$3! \times \left(\frac{4}{52}\right)^3$

h two are sevens and one is a six,

$3 \times \left(\frac{4}{52}\right)^2 \times \left(\frac{4}{52}\right)$

i exactly one is a diamond,

$3 \times \left(\frac{13}{52}\right) \times \left(\frac{39}{52}\right)^2$

j at least two of them are diamonds.

$3 \times \left(\frac{13}{52}\right)^2 \times \left(\frac{39}{52}\right) + \left(\frac{13}{52}\right)^3$

16 Three boys and three girls are to sit in a row. Find the probability that:

a the boys and girls alternate,

$\frac{2 \times 3! \times 3!}{6!} = \frac{1}{10}$

b the boys sit together and the girls sit together,

$\frac{2 \times 3! \times 3!}{6!} = \frac{1}{10}$

c two specific girls sit next to one another.

$\frac{2 \times 5!}{6!} = \frac{1}{3}$

17 On a school trip, there are 5 distinct hotel rooms A, B, C, D, E and 3 students X, Y, Z.
Find how many ways there are to fill the rooms by:

a Using permutations for ordered selection.

Ordered selection of 3 rooms from 5:

$5P3 = 60$

b Using combinations, then arranging.

Choose 3 hotel rooms from 5 to host students, then arrange the students.

$5C3 \times 3! = 60$

18 A parade has 3 music floats and 6 non-music floats.

How many ways are there of arranging the floats so none of the music floats are adjacent to each other?

Follow the steps to solve this problem.

Arrange non-music floats such that there are gaps: … N … N … N … N … N … N …

Ways to arrange 6 non-music floats: ……

$6!$

There are …… gaps. Ways to choose 3 positions for the music floats: ……

$7C3$

Ways to arrange the 3 music floats: ……

$3!$

Total: ……

$6! \times 3! \times 7C3 = 151\,200$

19 A family of five are seated in a row at the cinema. Find the probability that:

a the parents sit on the end and the three children are in the middle,

Parents: 2 positions, 2! ways
Children: 3! ways
$\frac{2! \times 3!}{5!} = \frac{1}{10}$

b the parents sit next to one another.

Treat parents as unit: 4! arrangements
Parents within unit: 2! arrangements
$\frac{4! \times 2!}{5!} = \frac{2}{5}$

20 Six people, of whom Patrick and Jessica are two, arrange themselves in a row.
Find the probability that:

a Patrick and Jessica occupy the end positions,

P and J on ends: 2! ways
Others: 4! ways
$\frac{2! \times 4!}{6!} = \frac{1}{15}$

b Patrick and Jessica are not next to each other.

P(not next to each other) = 1 - P(together)
$P(\text{not together}) = 1 - \frac{5! \times 2!}{6!} = \frac{2}{3}$

21 The letters of PROMISE are arranged randomly in a row. Find the probability that:

a the word starts with R and ends with S,

Fix R and S, need to arrange 5 middle letters
$\frac{5!}{7!} = \frac{1}{42}$

b the letters P and R are next to one another,

Treat PR as unit: 6! arrangements
PR or RP: 2!
$\frac{6! \times 2!}{7!} = \frac{2}{7}$

c the letters P and R are separated by at least three letters,

no letters between: 6! 2!
1 letter between: (5)5! 2!
2 letters between: (5P2) 4! 2!
$1 - \frac{6!\,2! + 5(5!\,2!) + (5P2)\,4!\,2!}{7!} = \frac{2}{7}$

d the vowels and the consonants alternate,

Only CVCVCVC is possible with 3 vowels
$\frac{4!\,3!}{7!} = \frac{1}{35}$

e the vowels are together.

Vowels OIE together as unit, arrange 5 units 5!
internal arrangements 3!
$\frac{5! \times 3!}{7!} = \frac{1}{7}$

22 The digits 3, 3, 4, 4, 4 and 5 are placed in a row to form a six-digit number.
If one of these numbers is selected at random, find the probability that:

a it is even,

Total: $\frac{6!}{2! \times 3!} = 60$
Even means it must end in 4
$\frac{\left(\frac{5!}{2! \times 2!}\right)}{60} = \frac{1}{2}$

b it ends in 5,

$\frac{\left(\frac{5!}{2! \times 3!}\right)}{60} = 1/6$

c the 4s occur together,

Treat three 4s as unit. 4 units, with repeated 3
$\frac{4!\,2!}{60} = \frac{1}{5}$

d the number starts with 5 and then the 4s and 3s alternate,

5 4 3 4 3 4
There is only one way to arrange this.
$\frac{1}{60}$

e the 3s are separated by at least one other number.

1 - P(3s together)
$1 - \frac{\left(\frac{5!}{3!}\right)}{60} = \frac{2}{3}$

23 The letters of KETTLE are arranged randomly in a row. Find the probability that:

a the two letters E are together,

Total: $\frac{6!}{2! \times 2!} = 180$
Treat EE as 1 unit
$\frac{\left(\frac{5!}{2!}\right)}{180} = \frac{1}{3}$

b the two letters E are not together,

$1 - \frac{1}{3} = \frac{2}{3}$

c the two letters E are together and the two letters T are together,

Treat EE and TT as units
$\frac{4!}{180} = \frac{2}{15}$

d the Es and Ts are together in one group.

Treat EETT as 1 unit.
3 units. $\frac{4!}{2!2!}$ internal arrangements
$\frac{3! \times \frac{4!}{2! \times 2!}}{180} = \frac{1}{5}$

24 A tank contains 20 tagged fish and 80 untagged fish. On each day, four fish are selected at random, and after noting whether they are tagged or untagged, they are returned to the tank. Answer the following questions, correct to three significant figures.

a What is the probability of selecting no tagged fish on a given day?

$\frac{{}^{80}C_4}{{}^{100}C_4} = 0.403$

b What is the probability of selecting at least one tagged fish on a given day?

$1 - 0.403 = 0.597$

c Calculate the probability of selecting no tagged fish on every day for a week.

$(0.403)^7 = 0.00174$

d What is the probability of selecting no tagged fish on exactly three of the seven days during the week?

${}^7C_3 \times (0.403)^3 \times (0.597)^4 = 0.291$

25 The letters A, E, I, P, Q and R are arranged in a circle. Find the probability that:

a the vowels are together,

All arrangements: (6-1)! = 5! = 120

Vowels (A,E,I): 3 letters
Treat vowels as one unit with consonants (P,Q,R): 4 objects
Circular: (4-1)! = 3! = 6
Internal vowel arrangements: 3! = 6
Total: 6(6) = 36

$P = \frac{36}{120} = \frac{3}{10}$

b A is opposite R,

Fix A, R has 1 possible opposite position.
Arrange remaining 4: 4! = 24

$P = \frac{24}{120} = \frac{1}{5}$

c the vowels and consonants alternate,

Vowels: A,E,I (3), Consonants: P,Q,R (3)

Fix one vowel, arrange 2 others in alternate positions: 2! = 2

Arrange 3 consonants: 3! = 6

Total: 2(6) = 12

$P = \frac{12}{120} = \frac{1}{10}$

d at least two vowels are next to one another.

Complement: no two vowels adjacent
(vowels and consonants alternate)

$P = 1 - \frac{1}{10} = \frac{9}{10}$

26 A committee of three women and seven men is to be seated randomly at a round table.

a What is the probability that the three women will sit together?

Treat 3 women as one unit with 7 men: 8 objects
Circular: (8-1)! = 7! = 5040
Internal women arrangements: 3! = 6
Total: 5040(6) = 30240

All arrangements: (10-1)! = 9! = 362880

$P = \frac{30240}{362880} = \frac{1}{12}$

b The committee elects a president and a vice-president. What is the probability that they are sitting opposite one another?

Fix president, vice-president opposite (1 of 9 positions)

$P = \frac{1}{9}$

27 There are two round tables, one oak and one mahogany, each with five seats. In how many ways may a group of ten people be seated?

Choose 5 for oak table: C(10,5) = 252

Arrange 5 on oak: (5-1)! = 4! = 24

Arrange 5 on mahogany: 4! = 24

Total: 252(24)(24) = 145 152

28 A bag contains seven white and five black discs. Three discs are chosen from the bag. Find the probability that all three discs are black, if the discs are chosen:

a without replacement,

$\frac{{}^5C_3}{{}^{12}C_3} = \frac{1}{22}$

b with replacement,

$\left(\frac{5}{12}\right)^3 = \frac{125}{1728}$

c so that after each draw the disc is replaced with one of the opposite colour.

A tree diagram may help you visualise this

$\frac{5}{12} \times \frac{4}{12} \times \frac{3}{12} = \frac{5}{144}$

29 Six people are to be divided into two groups, each with at least one person.

a Explain why the total possible ways to do this is 31.

Split two groups as (1,5): ${}^6C_1$
(2,3): ${}^6C_2$
(3,3): $\frac{{}^6C_3}{2}$
Total: 31

b Find the probability that there will be three in each group,

$\frac{\left(\frac{{}^6C_3}{2}\right)}{31} = \frac{10}{31}$

c Find the probability that there will be two in one group and four in the other,

$\frac{{}^6C_2}{31} = \frac{15}{31}$

d Find the probability that there will be one group of five and an individual.

$\frac{{}^6C_1}{31} = \frac{6}{31}$

30 A three-digit number is formed from the digits 3, 4, 5, 6 and 7 (no repetitions allowed). Find the probability that:

a the number is 473,

Total 3-digit numbers = 5P3 = 5 × 4 × 3 = 60
$\frac{1}{60}$

b the number is odd,

Odd last digit 3, 5, or 7
3 ways for last digit, then 4P2 ways to fill other 2 digits
$\frac{3 \times 4 \times 3}{60} = \frac{3}{5}$

c the number is divisible by 5,

ends in 5, 4P2 ways to fill other 2 digits
$\frac{4 \times 3}{60} = \frac{1}{5}$

d the number is divisible by 3,

Valid sets: {3,4,5}, {3,6,7}, {4,5,6}, {4,6,7}
Each set has 3! arrangements
$\frac{3! \times 4}{60} = \frac{2}{5}$

e the number starts with 4 and ends with 7,

3 choices for middle number
$\frac{3}{60} = \frac{1}{20}$

f the number contains the digit 3,

subtract cases where there is no 3
$1 - \frac{4 \times 3 \times 2}{60} = \frac{3}{5}$

g the number contains the digits 3 and 5,

Choose 3 positions for digit 3, choose 2 positions for digit 5, then last digit can be filled with 3 possible digits
$\frac{3 \times 2 \times 3}{60} = \frac{3}{10}$

h the number contains the digits 3 or 5,

$P(3) + P(5) - P(3 \text{ and } 5)$
Using part **f** and **g**
$\frac{3}{5} + \frac{3}{5} - \frac{3}{10} = \frac{9}{10}$

e all digits in the number are odd,

can arrange 3, 5, 7 in 3! ways
$\frac{3!}{60} = \frac{1}{10}$

f the number is greater than 500.

3 possibilities for first digit, then 4P2 ways to fill other two
$\frac{3 \times 4 \times 3}{60} = \frac{3}{5}$

31 A five-digit number is chosen at random. Find the probability:

a the number contains at least one four

Total 5-digit numbers: $9 \times 10^4 = 90\,000$
Without 4: 8×9^4
$1 - \frac{8 \times 9^4}{90000} = \frac{521}{1250}$

b the number contains at least one four and at least one five

No 4 and no 5: 7×8^4

$1 - \frac{7 \times 8^4}{90000} = \frac{856}{5625}$

c the number contains exactly three sevens

Place 3 sevens in 5 positions: 5C3
Fill other 2 spots with 9 possible digits: 9^2
But we need to subtract cases where the first digit is 0
First position is 0
Place 3 sevens in 4 positions 4C3
Fill other position with 9 possible digits: 9
$\frac{5C3 \times 9^2 - 4C3 \times 9}{90\,000} = \frac{43}{5000}$

d the number contains at least three sevens.

Need 3, 4, or 5 sevens

From part **c**, 3 sevens is $5C3 \times 9^2 - 4C3 \times 9$
Similarly, 4 sevens is $5C4 \times 9 - 4C4$
Only 1 way to have 5 sevens

$\therefore \frac{91}{10\,000}$

32 A survey has eight 'Yes/No'-type questions.
How many ways can the survey be answered if there are

a no restrictions?

$2^8 = 256$

b exactly three 'No' answers?

${}^8C_3 = 56$

c at least two 'No' answers?

Total - (0 'No') - (1 'No')
$= 2^8 - {}^8C_0 - {}^8C_1$
247

d at least six 'Yes' answers?

(6 'Yes') + (7 'Yes') + (8 'Yes')
$= {}^8C_2 + {}^8C_1 + {}^8C_0$
$= 37$

e at most two 'No' answers?

This is the same as 'at least six yes': 37

f at most six 'Yes' answers?

This is the same as at least two 'no' = 247

33 How many five-letter codes can be formed from the letters of the word COFFEE.

Consider cases: Exclude C, Exclude O, Exclude F, Exclude E

$\frac{5!}{2!\,2!} + \frac{5!}{2!\,2!} + \frac{5!}{2!} + \frac{5!}{2!} = 180$

34 A poker hand of five cards is dealt from a standard pack of 52.
Find the probability of obtaining the following:

a one pair,

Total number of ways: ${}^{52}C_5$
Choose the rank for the pair 13C1,
choose 2 cards from that rank 4C2,
choose 3 different ranks from remaining 12 ranks 12C3,
choose one card from each of those ranks: 4^3.

$$\frac{{}^{13}C_1 \times {}^{4}C_2 \times {}^{12}C_3 \times 4^3}{{}^{52}C_5} = \frac{352}{833}$$

b two pairs,

Choose 2 ranks for the pairs 13C2,
choose 2 cards from each rank: $(4C2)^2$,
choose 1 rank from remaining 11 for the fifth card 11C1,
choose 1 card from that rank: 4C1.

$$\frac{198}{4165}$$

c three of a kind,

Choose the rank for the triple 13C1,
choose 3 cards from that rank 4C3,
choose 2 different ranks from remaining 12C2,
choose one card from each: 4^2.

$$\frac{88}{4165}$$

d four of a kind,

Choose the rank for the four 13C1,
choose 4 cards from that rank 4C4,
choose another card from 48 remaining 48C1.

$$\frac{1}{4165}$$

e a full house
(one pair and three of a kind),

Choose rank for the triple 13C1,
choose 3 cards from that rank 4C3,
choose rank for the pair from remaining 12 ranks 12C1,
choose 2 cards from that rank 4C2.

$$\frac{6}{4165}$$

f a straight (five cards in sequence regardless of suit, ace high or low),

There are 10 possible sequences (A-2-3-4-5 through 10-J-Q-K-A).
For each sequence, choose a suit for each card: $4^5 = 1024$.

$$\frac{128}{32487}$$

g a flush (five cards of the same suit),

Choose a suit 4C1,
choose 5 cards from that suit 13C5.

$$\frac{33}{16660}$$

h a royal flush (ten, jack, queen, king and ace in a single suit).

There are only 4 royal flushes, one for each suit.

Alternatively, using combinations: choose a suit 4C1,

there is only one way to select 5 cards from those 5 ranks 5C5

$$\frac{1}{649\,740}$$

Exam Questions

1 2020 HSC Extension 1 Band 4

Out of 10 contestants, six are to be selected for the final round of a competition. Four of those six will be placed 1st, 2nd, 3rd and 4th.

In how many ways can this process be carried out?

A. $\frac{10!}{6!4!}$

B. $\frac{10!}{6!}$

C. $\frac{10!}{4!2!}$

D. $\frac{10!}{4!4!}$

C

$10C6 \times 6P4 = \frac{10!}{6!\,4!} \times \frac{6!}{2!} = \frac{10!}{4!\,2!}$

2 HSC Sample Question Extension 1 Band 4

Four girls and four boys are to be seated around a circular table. In how many ways can the eight children be seated if:

a there are no restrictions?

b the two tallest boys must not be seated next to each other?

c the two youngest children sit together?

a) $(8-1)! = 7!$

b) Fix tallest boy in a seat: 1

2nd tallest: 5 possible non-adjacent seats.

Arrangements of 6 other children: 6!

$5 \times 6!$

Alternatively,

Total – cases where they sit together

Consider two tallest as 1 unit, arrange 7 units around a circle and arrange internally

$7! - (7-1)! \times 2!$

c) Fix youngest in a seat: 1

2nd youngest: 2 possible adjacent seats

Arrangements of 6 other children: 6!

$2 \times 6!$

Alternatively, consider two youngest as a group

$(7-1)! \times 2!$

3 2021 HSC Extension 1 Band 4

A committee containing 5 men and 3 women is to be formed from a group of 10 men and 8 women.

In how many different ways can the committee be formed?

$10C5 \times 8C3 = 14\,112$

4 HSC Sample Question Extension 1 Band 4

A table tennis club consists of 6 males and 5 females.

How many committees of 4 players can be chosen that contain no more than 2 females?

Consider cases: 0 females, 1 female 2 females

6C4 + (5C1)(6C3) + (5C2)(6C2) = 265

5 2019 HSC Extension 1 Band 5

In how many ways can all the letters of the word PARALLEL be placed in a line with 3 Ls together?

A. $\frac{6!}{2!}$

B. $\frac{6!}{2!3!}$

C. $\frac{8!}{2!}$

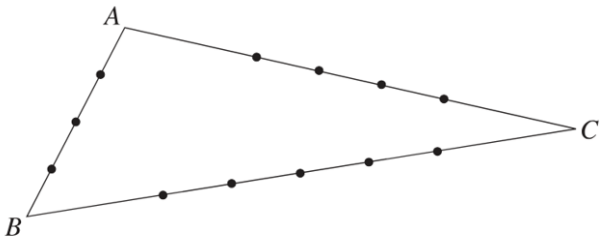
D. $\frac{8!}{2!3!}$

A

1Treat 3Ls as a single unit, 6 units total, with 2 repeated As

$\frac{6!}{2!}$

6 2022 HSC Extension 1 Band 5



How many smaller triangles can be formed using these shown points as vertices?

Can't take 3 points from a single side. Consider cases below:

Take 1 point from each side

$3 \times 4 \times 5 = 60$

Take 2 points from one side, and 1 point from the others.

Consider 2 points from AB, 2 points from AC, 2 points from BC

$3C2 \times 9C1 + 4C2 \times 8C1 + 5C2 \times 7C1 = 145$

Total = 60 + 145 = 205

Alternatively: Do Total – 3 from one side

$12C3 - 3C3 - 4C3 - 5C3 = 205$

7 HSC Sample Question Extension 1 Band 5

a In how many ways can the numbers 9, 8, 7, 6, 5, 4 be arranged in a circle?

$(6-1)! = 5! = 120$

b How many of these arrangements have at least 2 odd numbers together?

Total – no odd together

Consider the pattern OEOEOE

First arrange 3 odd numbers in a circle $(3-1)! = 2!$

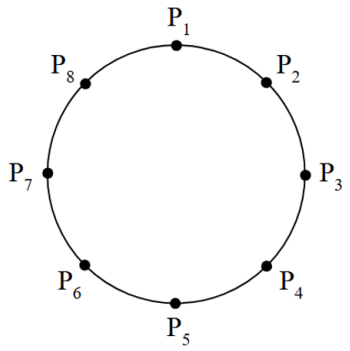
Arrange 3 odd numbers in the 3 gaps 3!

$2! \times 3! = 12$ ways

$120 - 6 = 108$

8 HSC Sample Question Extension 1 Band 5

Eight points are arranged in order around a circle.



a How many triangles can be drawn using these points as vertices?

$8C3 = 56$

b How many pairs of triangles can be drawn, where the vertices of each triangle are distinct?

Choose 3 from 8, then another 3 from 5 remaining

Divide by 2 as the triangle pairs are indistinguishable

$$\frac{8C3 \times 5C3}{2} = 280$$

9 HSC Sample Question Extension 1 Band 5

a In how many ways can the letters of COOKBOOK be arranged in a line?

$$\frac{8!}{4!\,2!} = 840$$

b What is the probability that a random arrangement has four O's together?

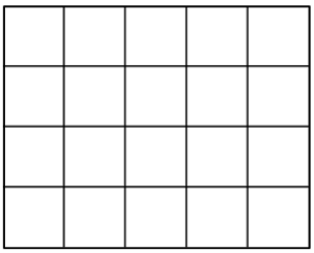
Treat 4 O's as 1 unit. 5 units, account for duplicate Ks

$$\frac{5!}{2!} = 60$$

$$\frac{60}{840} = \frac{1}{14}$$

10 HSC Sample Question Extension 1 Band 5

How many rectangles, including squares, can be found in the 4 x 5 grid below?



Squares/rectangles are made from 2 horizontal and 2 vertical sides

Choose 2 horizontal from 5, 2 vertical from 6

$5C2 \times 6C2 = 150$

Challenge Exercise

1 Eight students are trying out for a relay race team with four positions: first leg, second leg, third leg, anchor. Two of the students, Marcus and Sophie, cannot both be on the team.
How many ways are there of selecting the team?

a Solve this by considering the cases:
Case 1: Marcus on team, Sophie not;
Case 2: Sophie on team, Marcus not;
Case 3: neither on team.

Case 1:
Position for Marcus: 4 choices
Fill 3 remaining positions from 6 others: ${}^6P_3$
Total: $4 \times {}^6P_3 = 480$

Case 2:
Position for Sophie: 4 choices
Fill 3 remaining positions from 6 others: ${}^6P_3$
Total: $4 \times {}^6P_3 = 480$

Case 3: Neither on team
Fill 4 positions from 6 others: ${}^6P_4 = 360$

Total: 480 + 480 + 360 = 1320

b Solve this by finding the total, then subtracting the cases where they are both on the team:

i Using combinations.

Unacceptable ways: Fix Marcus and Sophie in the team, choose two others $6C2$, then find arrangements 4!

$8P4 - 6C2 \times 4! = 1320$

ii Without using combinations.

Unacceptable ways: choose 1 of 4 spots for Marcus, choose 1 of 3 spots for Sophie, then fill 2 remaining spots from 6 others

$8P4 - (4 \times 3 \times 6P2) = 1320$

2 How many 4-digit numbers can be formed using the digits 1, 2, 3, 4, 5, 6 (without repetition) such that the number is odd and greater than 4000?

Jimmy says: “I will find how many are greater than 4000, then divide by 2.”
Explain why he is incorrect and find the correct answer.

Among numbers > 4000, there is not an equal number of odd and even numbers.
Consider when the first digit is 4, the last digit can be 1, 3, 5 (3 odd possibilities)
When the first digit is 5, last digit can be 1, 3 (2 odd possibilities)
When the first digit is 6, last digit can be 4, 5, 6 (3 odd possibilities)

First digit 4: $3 \times {}^4P_2 = 36$
First digit 5: $2 \times {}^4P_2 = 24$
First digit 6: $3 \times {}^4P_2 = 36$
Total = 96

3 There are five backpackers and four available rooms in a hotel. How many ways can the backpackers be accommodated in the four rooms if each room can hold at most four people?

Each backpacker chooses any of 4 rooms
Subtract invalid cases (all 5 in one room)
Ways = $4^5 - 4$

4 Find how many arrangements there are of the letters of the word GUMTREE if:

a there are no restrictions

$\frac{7!}{2!} = 2520$

b the Es are together

$6! = 720$

c the Es are separated by 1 letter

Treat (E_E) as 1 unit, 5 units total
5 choices for the gap
$5 \times 5! = 600$

d the Es are separated by 2 letters

Treat (E_ _E) as 1 unit, 4 units total
5P2 choices for the gaps
$4! \times 5P2 = 480$

e the Es are separated by 3 letters

Treat (E_ _ _E) as 1 unit, 3 units total
5P3 choices for the gaps
$3! \times 5P3 = 360$

f the Es are separated by 4 letters

Treat (E_ _ _ _E) as 1 unit, 2 units total
5P4 choices for the gaps
$2! \times 5P4 = 240$

g the Es are separated by 5 letters

This is the same as fixing Es at the beginning and end.
5! ways to fill the middle letters = 120

h the G is somewhere between the two Es

Consider part c: Fix G as one in the gap 5! = 120
Consider part d: 2 places for G, then 4 other letters to fill other spot
$4! \times 2 \times 4 = 192$
Consider part e: 3 places for G, then 4P2 to fill other spots
$3! \times 3 \times 4P2 = 216$
Consider part f: 4 places for G, then 4P3 to fill other spots
$2! \times 4 \times 4P3 = 192$
Consider part g, G will always be in the middle if E is at the ends
120
120 + 192 + 216 + 192 +120 = 840

Alternatively, consider only the letters {E, E, G}
All arrangements patterns are:
*E*E*G*
*E*G*E*
*G*E*E*
So G is in between one third of the time (stars represent other letters)
$\frac{2520}{3} = 840$

Alternatively, consider that the pattern must be *G*E*E*
Choose 3 positions from 7 for GEE (in that order): 7C3
Then arrange the 4 other letters 4!
$7C3 \times 4! = 840$

i the M is somewhere to the left of both Es and the U is somewhere between them

all 12 possible relative orderings of {M, U, E, E}:

MEUE, MEEU, MUEE, EMUE, EMEU, EUME, EEMU, EEUM, UEME, UEEM, UMEE, UEEM

The condition will be satisfied $\frac{1}{12}$ of the time.
$\frac{2520}{12} = 210$

Alternatively, consider that the pattern must be MEUE
Choose 4 positions from 7 for MEUE in that order: 7C4
Then arrange 3 other letters: 3!
$7C4 \times 3! = 210$

j the G is somewhere to the left of the U and the M is somewhere to the right of the U

Consider relative orderings of {G, U, M}
GUM, GMU, UMG, UGM, MGU, MUG

Condition is satisfied $\frac{1}{6}$ of the time.
$\frac{2520}{6} = 2420$

Alternatively, consider that the pattern must be GUM
Choose 3 positions from 7 for GUM in that order: 7C4
Then arrange 4 other letters, where there are 2 Es: $\frac{4!}{2!}$

$7C4 \times \frac{4!}{2!} = 420$

5 If the letters of the word GUMTREE and the letters of the word KOALA are combined and arranged into a single twelve-letter word, in how many of these arrangements do the letters of KOALA appear in their correct order, but not necessarily together?

Choose 5 positions for {K, O, A, L, A} in that order 12C5

Then fill other 7 positions, remembering to account for 2Es

$12C5 \times \frac{7!}{2!} = 19\,995\,840$

6 Consider the digits 1, 2, 3, 4 and 5.

a How many four-digit passwords can be formed if there is no repetition?

5 × 4 × 3 × 2 = 5P4

b How many five-digit passwords can be formed, if there is no repetition?

5 × 4 × 3 × 2 × 1 = 5P5

c Explain why the answers from **a** and **b** are the same, even though there are differing numbers of digits.

Using the identity ${}^{n}P_{n} = {}^{n}P_{n-1} = n!$

Combinatorial argument: for every answer in **b**, simply delete the last digit and you have a valid answer for **a**

7

a Assuming a 365-day year, find the probability that in a group of three people there will be at least one birthday in common. Answer correct to two significant figures.

Total number of ways: 365^3

At least 1 in common: 1 - P(all different)

$1 - \frac{365 \times 364 \times 363}{365^3} = 0.0082$

b If there are n people in the group, find an expression for the probability of at least one common birthday.

$1 - \frac{{}^{365}P_n}{365^n}$

c Using Desmos plot a graph of the probability of at least one common birthday $P = \{\text{your expression from above}\}$ against number of people, x.
To plot permutations in Desmos: nPr(a,b).
Zoom in to the graph so that $0 \leq x \leq 60$ and $0 \leq P \leq 1$

https://www.desmos.com/calculator/v7ul5mtndg

d How many people need to be in the group before the probability exceeds 0.5?

23

e How many people need to be in the group before the probability exceeds 90%?

41

4 A room contains $n + 1$ people named P_1, P_2, P_3, ..., P_{n+1}, arranged in a line in that order. Person P_1 shakes hands with everybody else in the line and then leaves the room, having shaken everybody's hand. Afterwards, P_2 shakes hands with the remaining people in the room and similarly leaves. This is repeated until eventually, everybody has shaken hands with everybody else exactly once.

a How many hands did P_1 shake before leaving?

n hands

b When it was their turn, how many people did P_2 shake hands with?

$n - 1$ people

c How many hands did P_n and P_{n+1} shake?

1 and 0

d Explain why there were ${}^{n+1}C_2$ handshakes in total.

Each person shakes with every other person exactly once.

Number of ways to choose 2 people from n + 1 people = ${}^{n+1}C_2$

e Hence prove

$$1 + 2 + 3 + 4 + 5 \ldots + n = \frac{n}{2}(n + 1)$$

LHS represents total number of handshakes

Total number of handshakes is also ${}^{n+1}C_2$

${}^{n+1}C_2 = \frac{(n+1)!}{2!\,(n-1)!} = \frac{(n+1)(n)(n-1)!}{2!\,(n+1)!} = \frac{n(n+1)}{2}$

Therefore, LHS and RHS are ways to count total number of handshakes in this scenario, and so are equal.

5 A group of k people is to be selected from a total of $n+1$ people, then placed on k seats in a line.

a How many ways can this be done?

$^{n+1}P_k$

b Suppose Jimmy is one of the $n+1$ people. How many ordered selections are possible if he is excluded?

$^{n}P_k$

c How many ordered selections are possible if Jimmy is included?
Do this by fixing Jimmy first in the row, arranging the rest,
then multiplying by how many positions Jimmy can be in.

$^{n}P_{k-1} \times k$

d Hence deduce that $^{n+1}P_k = {}^{n}P_k + k(\,{}^{n}P_{k-1})$

The total number of ordered selections from n+1 people (part a) can be split into two cases:
one where Jimmy isn't in the group and the other where Jimmy is.

6

a Consider a sequence of four A's and six B's arranged randomly.
Using factorials, write down the number of ways of arranging the letters in the sequence.

$\frac{10!}{4!\,6!}$

b Express this as a binomial expression in the form $^{n}C_k$.

$^{10}C_4 = {}^{10}C_6$

c Now consider a sequence of three A's and four B's arranged randomly.
Write down the number of possible arrangements, in terms of $^{n}C_k$.

$\frac{7!}{3!\,4!} = {}^{7}C_3 = {}^{7}C_4$

d Explain why **a** and **c**, which are clearly permutation questions, can be answered using an expression normally used for combinations only.

Consider part **a**

Think of arranging four A's and six B's as making 10 independent binary decisions:
For position 1: choose A or B?, For position 2: choose A or B? … For position 10: choose A or B?

The constraint is that we must choose exactly 4 As and 6 Bs

We are essentially asking: out of 10 positions, how can I place 4 As? (the remaining positions must get Bs)

$\therefore {}^{10}C_4$

e A coin is flipped twenty times. How many ways can there be exactly twelve tails?

This is the same as arranging As and Bs, this time the binary choice is between Hs and Ts

$^{10}C_{12}$

7 Prove that

$$(1 \cdot {}^{n}C_1) + (2 \cdot {}^{n}C_2) + (3 \cdot {}^{n}C_3) + (4 \cdot {}^{n}C_4) + \cdots + (n \cdot {}^{n}C_n) = n \cdot 2^{n-1}$$

By considering the scenario:
from n people, choose a non-empty committee and choose a leader for the committee.

RHS:
Choose any person from n to be leader: n ways

Each of the remaining $n - 1$ people can either be on the committee or not: 2^{n-1} ways

Total: $n \cdot 2^{n-1}$

LHS:
For a committee of 1 person: choose 1 person from n, then choose 1 person out of 1 to be a leader: $1 \cdot {}^{n}C_1$

For a committee of 2 people: choose 2 people from n, then choose 1 person out of 2 to be a leader: $2 \cdot {}^{n}C_2$

For a committee of 3 people: choose 3 people from n, then choose 1 person out of 3 to be a leader: $3 \cdot {}^{n}C_3$

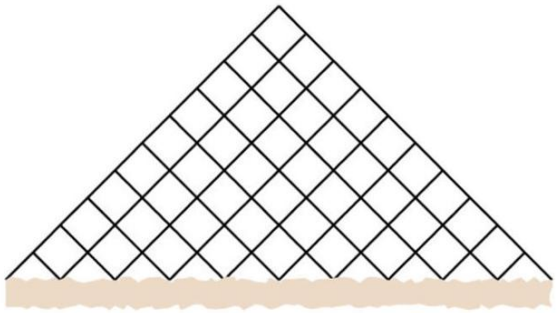
Continue until a committee of n people …

For a committee of n people: choose n people from n, then choose 1 person out of n to be a leader: $n \cdot {}^{n}C_n$

Total: $(1 \cdot {}^{n}C_1) + (2 \cdot {}^{n}C_2) + (3 \cdot {}^{n}C_3) + (4 \cdot {}^{n}C_4) + \cdots + (n \cdot {}^{n}C_n)$

Therefore, proved.

8 A drop of water is placed at the top of the grid. The drop must run downwards along edges of the grid. How many different ways can it reach the ground?



The water droplet must make a sequence of 10 moves to get to the ground.

Each move can either be left or right.

2^{10}

9 Suppose there are 6 identical lollies and 3 students, A B C. The lollies are to be distributed amongst the students. Not all students need to be given lollies. The stars and bars below show a way to distribute the lollies such that A gets 2, B gets 1, C gets 3.



a How many ways are there of distributing the lollies amongst the 3 students?

This is equivalent to arranging 6 stars and 2 bars

$\frac{8!}{6!\,2!} = 28$

Alternatively, choose 2 places out of 8 positions to put the two bars. $8C2 = 28$

b Hence prove that there are ${}^{(n+r-1)}C_n$ many ways to distribute n items amongst r people.

Distribute n items amongst r people.

This means n stars, $r-1$ bars.

Ways to choose where to put $r-1$ bars out of $n+r-1$ positions:

${}^{(n+r-1)}C_{(r-1)}$

$= {}^{(n+r-1)}C_n$ by symmetric property